What a Closed Agent's API Reveals: Estimating Its Hidden-State Geometry to Forecast and Prevent Violations

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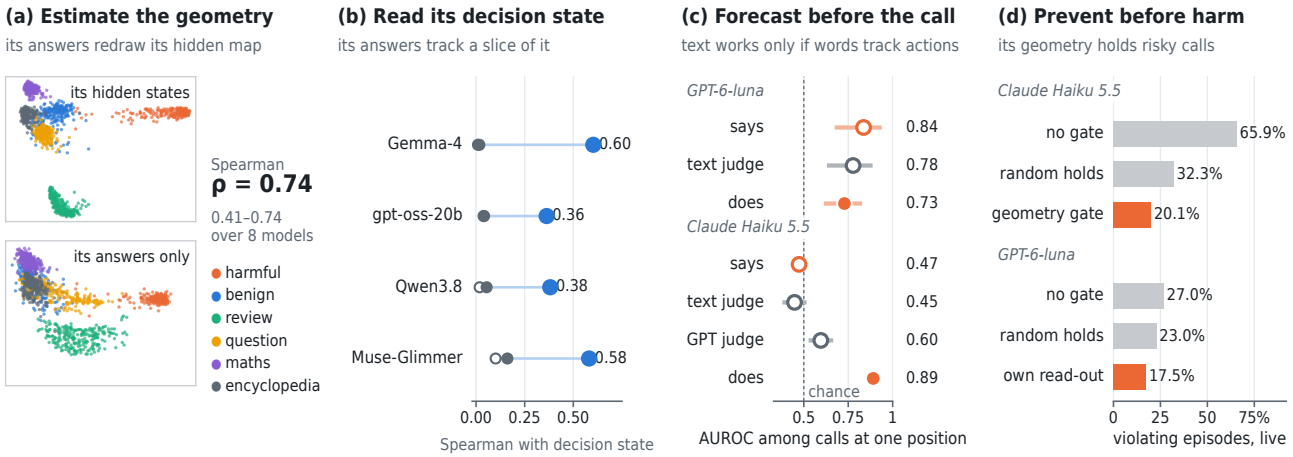


Figure 1. **Reading closed agents through their own APIs, with no training.** (a) Llama-3.1 8B's hidden geometry on 1,440 held-out contexts (classical MDS), and the same contexts placed only by its yes/no answers in exact Fisher–Rao coordinates ($\rho = 0.74$ for this model; 0.41–0.74 over 8 models). (b) At open agents' proposed calls, the 8 policy answers against the agent's decision state: answers predicted linearly from the state, across scenarios (blue; no label), the full state's geometry (grey), random rank-8 maps (hollow); secondary. (c) Forecast before the call (AUROC among calls at the same step): "says", the agent's answers about its next action; "text judge", the same model reasoning over the transcript ("GPT judge": GPT-6-luna, for Claude); "does", its own sampled next actions read by its own answers. Claude rows: restarts, its one task family with varying violations, the same 1,587 calls. (d) Live, episodes with an executed violation, with no reviewer: Claude, a gate on the geometry alone against random holds at the same dose (no-gate rate from a separate set of fresh units); GPT-6-luna, its own read-out against random holds. All from preregistered tests; Claude's rows in (c) replicated on fresh episodes.

Abstract

Probes on an agent's hidden states can stop an unsafe action before it runs, but they need the weights, and the most capable agents are closed; their words are no substitute, since agents that violate a policy often deny it. We estimate a closed agent's hidden-state geometry through its API alone, with no training, and use it to forecast and prevent violations. No API reveals hidden coordinates (a gauge symmetry), but their geometry is identifiable: yes/no answers sampled in throw-away copies of the conversation give exact Fisher–Rao coordinates. On eight open models it tracks the hidden geometry (Spearman 0.40–0.74); sampled answers reveal the part models share, log-probabilities the private part. On four open agents, the 8 policy answers are linearly predictable from the decision state across scenarios (0.36–0.60), while their geometry agrees with the full state at only 0.01–0.16. Read on the agent's sampled next actions, this geometry forecasts violations with no label and no question sign on Claude Haiku 5.5 (0.89; 0.87 on fresh episodes), weakly on GPT-6-luna (0.64), and on average over four open agents (0.75; 3 of 4 above chance). Claude's own words do not (GPT-6-luna's do): before 93% of its violating calls, none of its 32 answers admitted a violation. A probe on the true hidden state forecasts better (by 0.16, on two agents); on one agent, much of that lead may come from reading the agent's private reasoning. Live, on Claude's one task family with varying violations and with no human, a gate on the estimated geometry alone (no label, no question sign) left 20% of episodes

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violating, against 32% for random holds at the same dose and 66% with no gate (a first version, which broke ties in the cut at random, did not beat random holds). We report every preregistered test, passed or failed.

1. Introduction

Interpretability reads what a model knows and what it will do from its hidden state (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025). For agents, a probe on the hidden state can stop an unsafe action before it runs (Fayyazi et al., 2026). These tools need the weights. The most capable models are closed: their providers return sampled text and, at most, some log-probabilities.

Why an outside read of the state. Agents make calls that they are not authorized to make, and their words cannot be trusted to say so: in CAPE, four open frontier agents that knew a call was unauthorized made it and then denied it (Fayyazi et al., 2026), and chain-of-thought often leaves out what drives an answer (Turpin et al., 2023; Chen et al., 2025). A monitor needs the agent's state, read from outside its own account. For a closed agent the only outside access is its API.

What can an API reveal of the hidden state? Not hidden coordinates: any change of basis that the next layers undo leaves every output unchanged (Proposition 1). Geometry survives. We ask the model yes/no questions in *side branches* (Figure 2): copies of the conversation, each with one question appended, sent through its API and then discarded, so the agent never sees them. In the angles $\vartheta = 2 \arcsin \sqrt{P(\text{YES})}$, the Fisher–Rao distance between answer laws is Euclidean, its noise is known, and near a state it is the hidden geometry seen through the metric that the questions pull back (Theorem 1). Nothing is trained.

What we find (Figure 1). The theory has three steps, and each makes a prediction that we test.

1. **The geometry is estimable through the API** (Section 3). On eight open models, the answer geometry tracks the hidden geometry (0.41–0.74), better than embeddings of the input or replies on 6–7 of 8. Sampled answers reveal the part of the geometry that models share (a model's answers match another model's hidden geometry as well as its own; excess +0.002), while the model-private part appears only in log-probabilities. 8 policy answers are linearly predictable from an agent's decision state across scenarios (0.36–0.60), though their geometry agrees with the full state at only 0.01–0.16 (secondary).
2. **The geometry predicts violations** (Sections 4.1 and 4.2). Where the transcript decides the violation (Corollary 1), the estimated geometry suffices at the call: a fixed projection of it ranks violations nearly as well as a probe on the hidden state (0.82–0.99 on four open agents, 0.98 on GPT-6-luna), and the geometry alone, with no label and no question sign, flags them. Before the call, the forecast is the agent's policy (Proposition 3), which its API supplies by sampling: the geometry of the agent's sampled next actions forecasts its violations with no label and no sign on Claude Haiku 5.5 (0.89, replicated at 0.87 on fresh episodes) and, on average, on four open agents across their task families (0.75; 3 of 4 above chance). Claude's words do not (GPT-6-luna's do): before 93% of Claude's violating calls, none of its 32 answers about its next action admitted a violation (descriptive). With CAPE's checker labelling the sampled calls, the theory predicted in advance the loss of another model's draws on both closed agents (a first Claude test failed). A white-box probe forecasts better (by 0.16 on two agents, registered); on Gemma-4 much of that lead may come from reading the agent's private reasoning, but our test was too small to show parity at the start of the turn.
3. **The geometry prevents violations** (Section 4.3). On Claude's restarts, with no human in the loop, a gate on the estimated geometry alone left 20% of episodes violating, against 32% for random holds at the same dose (a first version that broke ties at random did not beat them); the signed read-out also beat random holds (35% vs 43%), and holds stack on a policy reminder (21% to 11%); with CAPE's checker reviewing held calls, a law fixed in advance predicted five agents' reductions (an earlier form failed).

Contributions. A theory of what an API reveals and of monitoring versus forecasting (Section 2; mostly classical, its value in predictions fixed before the data); evidence on eight open models, six agents and three benchmarks; a released tool (Section H).

Preregistration. We hash-locked every confirmatory test before its data existed (or, for a test on existing data, before its analysis) and published each lock before the run; a result is preregistered unless marked exploratory. Of the method's 82 tests, 38 passed every primary endpoint, 26 passed some, 14 failed, 2 stopped early and 2 are pending; Section L lists every test with its outcome, and each claim below names the failed tests that bear on it. Analyses registered on existing data after related results were known are marked secondary; their positive Claude Haiku 5.5 results were retested on fresh episodes and replicated (their GPT-6-luna parts were not retested). Three headline tests redesign earlier tests that failed (the prevention law, the pre-call gate, Claude's forecast with exact contexts), reported next to them. Over 37 registered closed-agent endpoints through G64, 25 survive a Holm correction (exploratory); the later replications passed their own Holm families.

2. Theory

2.1. Setting

A language model reads a context x (tokens) and reaches an internal state $s(x)$, a deterministic function of the tokens. An outside observer sees only outputs: sampled text and, on some APIs, the log-probabilities of the next token. A *battery* $Q = (q_1, \dots, q_m)$ of yes/no questions is asked in *side branches* (Figure 2): each question is appended to a copy of x in its own call, the copy is then discarded, so the model's real conversation never sees it, and the answers are independent given x. The *observable state* is $\Psi_Q(x) = \big(P(\text{YES} \mid x \oplus q_j)\big)_j$, and the answers follow the product law $P_x = \bigotimes_j \text{Bernoulli}(\Psi_{q_j}(x))$. Where a model is open, its hidden state $h(x) \in \mathbb{R}^d$ at a layer is the ground truth that we test against.

2.2. What cannot be read: a gauge symmetry

Proposition 1 (gauge)**.** *Take a pre-norm residual network with RMS normalisation, with the normalisation gains folded into the adjacent weights. For every orthogonal G on the residual stream, the network whose embedding, write matrices and write biases are multiplied by G on the left, and whose read matrices and unembedding are multiplied by $G^\top$ on the right, has hidden states Gh and the same output distribution on every input. At a layer that is read only through linear maps, the same holds for every invertible G, with G^{-1} in place of $G^\top$.*

No estimator that uses only outputs can recover hidden *coordinates* beyond this group (computational invariance; Ashkboos et al., 2024). Distances and angles between states are invariant under the orthogonal group, so the *geometry* of the states is the natural target. Post-block normalisation (Gemma-3, OLMo-3) restricts G to the maps that commute with its gains (Section B).

2.3. What can be read: the observable geometry

Let $d_Q(x, x')$ be the Fisher–Rao geodesic distance between the answer laws P_x and $P_{x'}$.

Theorem 1 (observable geometry)**.** *(a)* Exact flat coordinates. *With* $\vartheta_j = 2 \arcsin \sqrt{\Psi_{q_j}}$, $d_Q(x, x')^2 = \sum_j \big(\vartheta_j(x) - \vartheta_j(x')\big)^2$. *(b)* Invariance and uniqueness. *d_Q depends on the outputs only, so it is gauge invariant; up to a constant, the Fisher metric is the only Riemannian metric on the simplex over $\{0,1\}^m$ invariant under Markov morphisms (Čencov), and d_Q is its geodesic distance on the independent answer laws. (c)* Relation to the hidden state. *If $\Psi_j = \sigma(a_j(s))$ with smooth a_j, then for nearby states $d_Q^2 = \Delta^\top G_Q(s)\,\Delta + O(|\Delta|^3)$, with the pulled-back Fisher metric $G_Q = \sum_j \Psi_j(1-\Psi_j)\nabla a_j \nabla a_j^\top$. (d)* Estimation from sampled answers. *Let $\mu_n(p) = \mathbb{E}\,\hat\vartheta$ for the Anscombe angle $\hat\vartheta = 2 \arcsin \sqrt{(k + 3/8)/(n + 3/4)}$, $k \sim \text{Bin}(n, p)$; μ_n is known, strictly increasing, and $\mu_n \to \vartheta$. For two independent halves A, B of n samples each, $\hat D^2(x, x') = \sum_j \big(\hat\vartheta_j^A(x) - \hat\vartheta_j^A(x')\big)\big(\hat\vartheta_j^B(x) - \hat\vartheta_j^B(x')\big)$ satisfies $\mathbb{E}\hat D^2 = \sum_j \big(\mu_n(\Psi_j(x)) - \mu_n(\Psi_j(x'))\big)^2$ exactly, for every Ψ, when every cell has the same n.*

Parts (a)–(c) are classical information geometry (Amari, 2016); part (d) is ours. Part (a) is a change of variables: nothing is fit, and classical scaling of d_Q gives coordinates that are exact up to a rigid motion. Part (b) says these are the only natural coordinates; Picozzi (2026) study the same object for next-token laws. Part (c) ties them to the hidden state: agreement should rise as more questions give G_Q more rank. Part (d) makes the geometry measurable on any chat API, at $m \cdot n$ one-token calls per context (exactness needs the same n in every cell and answers that are yes or no; Section B).

Checking a closed model's reconstruction. A closed model's hidden states cannot be seen, so how well its answer geometry matches its own hidden geometry cannot be measured directly. It can be estimated, because hidden geometries share a common part across models (Section 3).

Proposition 2 (triangulation)**.** *On the vector of all context-pair distances, let the hidden geometry of model m be $H_m = \lambda_m C + e_m$ and the answer geometry of answerer a be $A_a = \kappa_a C + f_a$ (standardized; the parts e_m, f_a uncorrelated with C and across models). Then $\rho(H_a, H_b) = \lambda_a \lambda_b$ and $\rho(A_a, H_b) = \kappa_a \lambda_b$ for $a \neq b$, and the unseen agreement is $\rho(A_c, H_c) = \kappa_c \lambda_c + \rho(f_c, e_c)$. Open models identify the λ_b; a closed model's answers against them identify κ_c.*

With no own-model excess ($\rho(f_c, e_c) = 0$) and λ_c in the open models' range, $\kappa_c \cdot \text{median}(\lambda)$ estimates the closed model's own agreement; leaving each open model out tests the estimator.

Where an API returns full log-probabilities, the final state's geometry is recovered exactly, not estimated (Theorem 2, Section B.2, after Carlini et al. (2024, App. H)); closed APIs do not return them, so the grey box does not use this.

2.4. From the geometry to monitoring agents

Setting. At step t of an episode the operator holds the visible tokens V_t (instructions, task, earlier calls and their results) and sees the proposed call A_t; $Y \in \{0, 1\}$ marks a policy violation. A monitor scores (V_t, A_t) and holds the call when the score passes a cut; its budget β is the share of calls it holds.

Corollary 1 (the transcript suffices)**.** *Let $Y = g(A_t, V_t, W)$, where the world facts W satisfy $W \perp\!\!\!\perp Z_t^- \mid V_t$ and Z_t^- are all tokens before the call, hidden ones included. Then $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$: no monitor of the agent's*

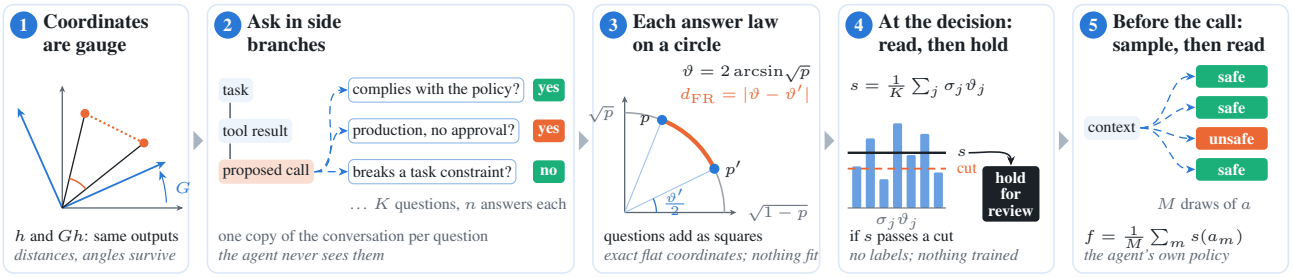


Figure 2. **The method, with no training and nothing but the API.** (1) Hidden coordinates are not observable (Proposition 1). (2) At a proposed call, the agent answers yes/no questions about the call in side branches of its own conversation, which the conversation never sees. (3) A Bernoulli law p is the point $(\sqrt{1-p}, \sqrt{p})$ on a quarter circle, and its Fisher–Rao distance to another is twice the arc between them; independent questions add as squares, so the angles are exact flat coordinates (Theorem 1). (4) The angles, signed by each question's wording, are averaged into a fixed score that holds the call for review before it runs. (5) Before the call exists, the agent's next action is drawn M times in side branches, and each drawn call is scored as in (4) or by its distance from the agent's typical state; the mean forecasts the violation (Proposition 3 and Corollary 2). Any capable reader can score the draws. Asked about any context, the same coordinates give the observable geometry of Section 3.

internals has a lower Bayes risk than the best monitor of (V_t, A_t).

The premise concerns the label, and we can test it. It holds when facts shown in the transcript decide the violation, as in CAPE-Bench and most τ^2 writes. It fails when a fact the agent never checked decides it. Then no monitor of the transcript or of the agent's internals can see the violation (Section 4.1). Where it holds, a monitor needs the transcript and the call, not the weights.

Proposition 3 (the forecast is the agent's policy)**.** *Before step t the agent acts by $A_t \sim \pi(\cdot \mid V_t, Z_t^-)$; let $\bar\pi(a \mid V_t)$ be this policy averaged over the hidden states consistent with V_t, and $r(a, V_t) = P(Y \mid a, V_t)$. The Bayes forecast is $f = \sum_a \bar\pi(a \mid V_t)\, r(a, V_t)$. A reader that forecasts with another policy π', $f' = \sum_a \pi'(a \mid V_t)\, r(a, V_t)$, has squared-loss excess risk*

$$\mathbb{E}(Y - f')^2 - \mathbb{E}(Y - f)^2 = \mathbb{E}\,\Delta^2 \le \mathbb{E}\,\|\bar\pi - \pi'\|_1^2,$$

with $\Delta = \sum_a (\bar\pi - \pi')(a \mid V_t)\, r(a, V_t)$. The excess is zero once the call A_t is visible.

Proof sketch. $f = \mathbb{E}[Y \mid V_t]$, so $\mathbb{E}(Y-f')^2 = \mathbb{E}(Y-f)^2 + \mathbb{E}(f-f')^2$; and $|f - f'| \le \|\bar\pi - \pi'\|_1$ since $0 \le r \le 1$. Given A_t, both forecasts equal $r(A_t, V_t)$. □

Corollary 2 (reading the policy by sampling)**.** *Let $a_1, \dots, a_M$ be next actions drawn from the agent in side branches, so $a_k \sim \bar\pi(\cdot \mid V_t)$ i.i.d., and $\hat f = \frac{1}{M}\sum_k r(a_k, V_t)$. Then $\mathbb{E}(Y - \hat f)^2 - \mathbb{E}(Y - f)^2 = \mathbb{E}\,\mathrm{Var}_{\bar\pi}[r(a, V_t)]/M \le 1/(4M)$.*

The corollary is elementary (a Monte Carlo mean); its point is that sampling the agent reads its own policy, and with it the Bayes forecast; a self-report or another model's draws are readers with another policy and pay the excess of Proposition 3. Since $\mathbb{E}(Y-q)^2 - \mathbb{E}(Y-f)^2 = \mathbb{E}(f-q)^2$ for any forecast q of the context alone, the draws estimate a text monitor's loss with no label: $\overline{(\hat f - q)^2} - \overline{\hat f(1-\hat f)}/(M-1)$. To forecast, r need not be calibrated: we replace it by the read-out of each drawn call, the agent's own answers at that call. To predict a loss, it must be (Section 4.2). So at the call any capable reader can match the agent (Corollary 1); before it, a reader loses exactly as much as its policy differs from the agent's on violating actions. Another model's answers can forecast well only where it would act like the agent (tested in Section 4.2).

The observable state at the decision. At a proposed call we ask the agent, through its own API, m yes/no questions about the call in side branches. Their angles $\vartheta_t \in [0, \pi]^m$ are its observable state, in the coordinates of Theorem 1. The score is a fixed read-out, the mean angle in each question's violation direction:

$$s_t = \frac{1}{m}\sum_{j=1}^{m} \sigma_j\, \vartheta_{tj}, \qquad \sigma_j \in \{-1, +1\}. \tag{1}$$

Here $\sigma_j = -1$ for the question "does it comply" and $+1$ for the others. The read-out is one projection of the observable geometry. The geometry alone also gives a monitor: a call's Fisher–Rao distance from the agent's typical state, $g_t = \|\vartheta_t - \tilde\vartheta\|$ with $\tilde\vartheta$ the median of the agent's own unlabelled calls (Theorem 1). It needs no sign: rewording a question to its negation maps ϑ_j to $\pi - \vartheta_j$ and leaves g_t unchanged. No label is used and nothing is fit. Self-policing is the one-question case: hold the call when the agent's own answers say that it does not comply.

Proposition 4 (a label-free ceiling)**.** *Let N be the sampling-noise variance of s_t and $\mathrm{SNR} = \mathrm{Var}(s_t)/N - 1$; neither uses a label. If the noise is independent of Y and s_t is Gaussian with equal variance in the two classes, then for a violation rate p, $\mathrm{AUROC}(s_t) \le \Phi\big(\sqrt{\mathrm{SNR}^+/(2p(1-p))}\big)$,*

with $\mathrm{SNR}^+ = \max(\mathrm{SNR}, 0)$*; at a true* $\mathrm{SNR} \le 0$ *the read-out cannot rank violations.*

Proof sketch. With class gap δ and within-class variance $\sigma_{\text{in}}^2 \ge N$, $\mathrm{Var}(s_t) = \sigma_{\text{in}}^2 + p(1-p)\delta^2$, so $p(1-p)\delta^2 \le \mathrm{SNR} \cdot N \le \mathrm{SNR} \cdot \sigma_{\text{in}}^2$; and $\mathrm{AUROC} = \Phi(\delta/\sqrt{2}\sigma_{\text{in}})$. □

In practice we use the nominal noise $\frac{1}{m^2}\sum_j (n_j + \frac{1}{2})^{-1}$ for n_j parsed answers; it overstates the noise near $P(\text{YES}) \in \{0, 1\}$, so the nominal SNR is conservative: a high value supports a setting, and a low value flags it without ruling it out. More answers lower the noise and raise the ceiling.

The cut. A cut on the read-out is set on the agent's own unlabelled traffic and then frozen (Proposition 7); a state-space model of the angles across steps is in Section C.

Metrics. For geometries: Spearman correlation between two distance matrices over all pairs of contexts (no fitted parameter). For monitors, we use two metrics. The first is the *within-stratum AUROC*: the AUROC over pairs of one violating and one safe call from the same stratum. A stratum is a template (a CAPE-Bench task family), a tool, or an AgentDojo suite; where stated, it also fixes the call's *position* (its ordinal among the episode's calls, also called its step). The second is the share of violations caught when exactly β of the actions are escalated. Intervals are 95% percentile intervals from a bootstrap over contexts, scenarios or episodes. Proofs are in Section B.

3. Estimating the hidden geometry through the API

We first test the reconstruction where the truth is visible: on open models, whose hidden states we store. **Data.** 4,800 contexts of six kinds (1,440 held out) and 1,000 fresh ones with four new kinds; 88 questions; eight open models from seven families (Section F). Ground truth: each model's residual stream at the last prompt token, at a layer chosen on development data (Section F).

Answers beat the input and the replies. On every model, the answer geometry matches the hidden geometry better than a sentence embedding of the input does (gains +0.16 to +0.49; every lower bound at least 0.141; Figure 3a, Table 8). The model's own replies, the other output that an API gives, do little better than the input (0.22–0.44). Agreement rises with more questions (+0.02 to +0.25); 8 sampled answers per question keep 97–100% of the exact agreement. On 1,000 fresh contexts (400 of them of four new kinds), answers beat the input on 6 of 8 models. Against the strongest of five text baselines (three sentence embedders, on the input or the replies), answers win on 7 of 8 models on held-out contexts (gpt-oss ties: +0.013 [−0.020, 0.041]) and on 6 of 8 on fresh ones (Gemma-3 and gpt-oss lose by 0.10 and 0.15; exploratory). Raw $P(\text{YES})$ ranks about as well as the exact angles; what the angles add is a known noise, which the read-out's label-free ceiling uses (Section 4.1).

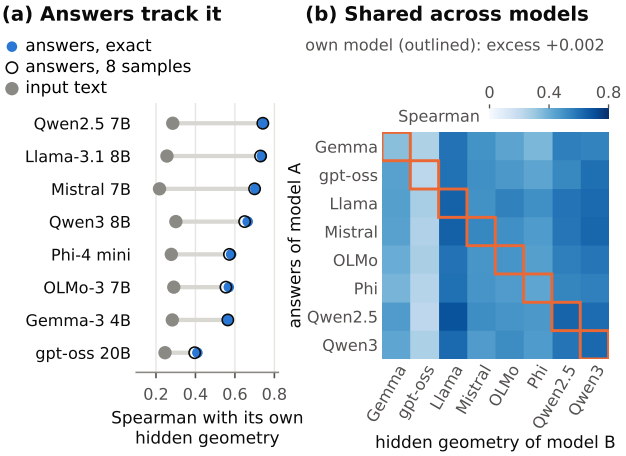


Figure 3. **What yes/no answers reveal of a hidden state.** (a) Agreement of each open model's answer geometry with its own hidden geometry, exact and from 8 sampled answers, against the geometry of the input text (Spearman over all pairs of 1,440 held-out contexts; nothing fit). (b) Answers of model A against the hidden geometry of model B on 1,000 fresh contexts: the own-model diagonal (outlined) is not special.

The geometry is shared across models. A model's answers match another model's hidden geometry exactly as well as its own. On 1,000 fresh contexts, the mean own-model excess is +0.002 [-0.001, +0.006], inside the equivalence band of ± 0.02 that we fixed in advance (Figure 3b). So what the answers reveal is the geometry that models share, in line with converging representations (Huh et al., 2024; Picozzi, 2026). It also predicts the next section's result at the call: the shared part is what any model's answers can read.

Which channel reveals what. On 1,000 further fresh contexts (preregistered), sampled yes/no answers carry no own-model excess (+0.001), while at the last layer the model-private part grows with the output channel (top-1 +0.056, top-20 +0.115, log-probabilities +0.221 [0.211, 0.230]); log-probabilities pick out the model in 7 of 8 cases, answers in 1. A closed API without log-probabilities therefore exposes the shared geometry of the state, and the agent's private part only through what it does (Section 4.2).

A short battery estimates a slice of the decision state. By Theorem 1(c), m questions estimate the hidden geometry in the metric they pull back, a slice of rank at most m, not the full state. At agents' proposed calls, 8 policy questions agree with the full decision-state geometry at only 0.01–0.16; a linear map from the decision state to the answers, fit across scenarios with no label, gives the slice they pull back, and the answer geometry agrees with those predictions at 0.60, 0.36, 0.38 and 0.58 on the four open agents, against

0.02–0.10 for random rank-8 maps of the state (all 4 lower bounds above both; secondary). The answers are a linear read-out of the decision state, not of its full geometry. 32 broader questions recover more of the full state (+0.12 [0.08, 0.16], still 0.18) but their geometry does not flag violations (0.50; policy battery 0.58–0.96).

A closed model reveals the shared geometry, on familiar kinds of context. Claude Haiku 5.5 answered 24 questions, 8 times each, on 192 held-out contexts. Two independent halves of its answers give the same geometry (Spearman 0.95 [0.94, 0.96]; bar 0.7). Its answers agree with open models' hidden geometries at 0.45–0.51. On 400 fresh contexts that include new kinds (news, code, jailbreaks, injections), a preregistered test failed: its 24 answers agreed with the open models' hidden geometries at only 0.10–0.48, and its own replies, embedded, scored higher on 7 of 8 models (significantly on 5). A closed model's general geometry is estimated well only on contexts like those its questions were written for; the agent results below use batteries written for the agent's task. With no own-model excess, triangulation (Proposition 2) estimates Claude's agreement with its own hidden geometry at 0.49 [0.44, 0.55] (leave-one-out error 0.061 [0.051, 0.070]; 0.114 on fresh contexts).

4. Monitoring, forecasting and preventing violations

Agents and benchmarks. CAPE-Bench (Fayyazi et al., 2026) gives matched tool-use tasks in which one fact decides whether an action is authorised; its checker labels each action. Open agents (decision states stored): Gemma-4 (26B), gpt-oss-20b, Qwen3.8-27B, Muse-Glimmer-30B. Closed agents, through their APIs, reasoning off: Claude Haiku 5.5, GPT-6-luna. External benchmarks: AgentDojo and Sierra's τ^2-bench airline.

4.1. Monitoring at the call

The read-out. At each proposed call, the agent answers 8 yes/no questions about it through its own API, 4 times each (Section F.1), scored by the fixed read-out (Equation (1); no labels).

It ranks violations about as well as a probe on the hidden states. On held-out calls, the within-template AUROC is Gemma-4 0.994, gpt-oss 0.822, Qwen3.8 0.951, Muse-Glimmer 0.929 (Table 1). A probe trained on the agent's true decision states, a comparison that needs the weights and labels, scores Gemma-4 0.980, gpt-oss 0.772, Qwen3.8 0.826, Muse-Glimmer 0.941; the mean difference is +0.045 [−0.017, 0.135]. The read-out beats the agent's single self-audit answer by +0.036 [0.006, 0.064]. On the open agents, it also beats a Gemma-4 judge that reasons, by +0.073 [0.021, 0.124]. On Claude Haiku 5.5, whose answers we take with reasoning off, the judge is better (Table 1). The read-out reads the slice of the state its questions pull back (Section 3), not the full state: on open agents about 20% [13, 27] of its ranking power runs through the hidden violation direction, the rest through the transcript, which any monitor can see.

Table 1. **Monitoring a decision through the agent's own answers** (held-out calls). AUROC within strata; best per row in blue; Claude = Claude Haiku 5.5. The read-out of 8 answers is fixed (nothing fit to labels). The trained probe reads the agent's true hidden states (32 principal components, fit on development calls); it is a comparison, not part of the method (more: Section D).

Agent	calls	**8 answers (ours)**	1 answer (self-audit)	judge[d] (reasoning)	trained probe	tuned[c] probe
OPEN AGENTS on CAPE-Bench (within template)						
Gemma-4[e]	600	**0.994**	0.966	0.989	0.980	0.991
gpt-oss-20b	599	0.822	0.758	0.755	0.772	**0.841**
Qwen3.8-27B	373	**0.951**	0.894	0.780	0.826	0.950
Muse-Glimmer-30B	600	0.929	0.935	0.880	0.941	**0.972**
CLOSED AGENTS on CAPE-Bench (no probe can exist)						
Claude, restarts[a]	149	0.909	0.323	**1.000**	–	–
Claude, all[b]	495	0.475	0.306	**0.693**	–	–
GPT-6-luna	560	**0.979**	0.795	–	–	–
AGENTDOJO (within suite) AND τ^2-BENCH (within tool)						
Gemma-4, injection	816	**0.985**	0.975	–	–	–
gpt-oss-20b, injection	821	**0.904**	0.900	–	–	–
Claude, airline writes	360	**0.704**	0.555	–	–	–

[a]A trained reader of the call text scores 1.000 here. [b]Pooled; mailing-list sends are decided by a fact it never checked. [c]Exploratory: components and regularisation chosen by cross-validation. [d]Gemma-4 with reasoning, one holistic probability per conversation. [e]New task families.

With no label, the read-out's SNR predicts where it works. This is where the geometry's coordinates pay off: their sampling noise is known (Theorem 1), so the read-out's variance beyond that noise, its SNR, needs no label (Proposition 4). Over 29 cells (agent, benchmark and stratum), the SNR predicts the AUROC (Spearman 0.71): cells with SNR > 0 score 0.96 on average, cells with SNR ≤ 0 score 0.68 (the nominal noise is conservative, so a low SNR flags a setting without ruling it out).

At the call, another answerer does as well, as the theory implies. The transcript decides the violation (Corollary 1) and the answers read a geometry that models share, so another model answering about the same call should match the agent: on open agents it does (own − other +0.012 [−0.018, 0.040]), and on Claude Haiku 5.5 it does better (−0.09 on restarts; −0.28 on τ^2 with reasoning on). Reasoning may help at the call: Claude reviewing its own τ^2 writes with reasoning on reaches 0.918 on 180 writes, against 0.699 for our read-out with reasoning off (difference not significant).

Other benchmarks and closed agents. On AgentDojo (Debenedetti et al., 2024), 8 questions written from the injection policy before any answer existed rank the open agents' injected calls at 0.985 [0.964, 0.998] and 0.904 [0.872, 0.932]. GPT-6-luna's answers rank its violations at 0.979 [0.946, 0.998]. Claude Haiku 5.5's rank its unapproved restarts at 0.909 [0.858, 0.952], but pooled score

Table 2. **Forecasting the coming violation before the call**, every forecaster on the same calls (AUROC; 95% interval where measured). Claude Haiku 5.5: unapproved production restarts with exact contexts, the original set and a prospective replication on fresh episodes (registered before any episode existed). "=": same model as the row above. *Exploratory (descriptive, not registered); the other model is Gemma-4, sampling from Claude's exact contexts.

Forecaster	Claude Haiku 5.5 original (1,587)	Claude Haiku 5.5 replication (1,582)	GPT-6-luna (1,134)
AMONG CALLS AT THE SAME STEP			
says: self-report	0.47 [0.44, 0.50]	0.47	**0.84** [0.67, 0.94]
text: the agent reasoning over its transcript	0.45 [0.38, 0.52]	0.55	0.78 [0.63, 0.89]
text: GPT-6-luna reasoning over it	0.60 [0.53, 0.67]	0.61	=
does: own sampled actions, own read	0.89 [0.85, 0.93]	0.87 [0.83, 0.91]	0.73 [0.61, 0.83]
does: same actions, read by GPT-6-luna	**0.93** [0.89, 0.96]	**0.92**	–
other: Gemma-4 sampling from its contexts*	0.54	–	–
does: same actions, geometry only	0.89 [0.85, 0.92]	0.87 [0.82, 0.91]	0.64 [0.54, 0.74]
ACROSS STEPS			
the step alone	0.82	0.83*	**0.78**
own sampled actions alone	0.80	0.79*	0.68
Gemma-4 sampling from its contexts*	0.82	–	–
best text monitor	0.90	0.92	**0.78**
best text monitor + own sampled actions	**0.96**	**0.96**	–

0.475: a fact it never checked decides its mailing-list sends; on τ^2-bench, 0.704 [0.536, 0.822].

4.2. Forecasting before the call

The pre-call state. Before each call, the agent answers 8 questions about its next action (its self-report); Table 2 compares every forecaster on the same calls, at the same step.

When text suffices: GPT-6-luna. Its words track its actions: its self-report (0.84) and a text monitor (0.78) forecast as well as its sampled actions (0.73), a reading made after the data (the registered opposite expectation failed).

When text does not suffice: Claude Haiku 5.5. Claude Haiku 5.5 rarely violates (2 of 24 CAPE-Bench templates; 1.5–1.6% on AgentDojo and τ^2), and on one template it violates nearly always; its tests use the other, unapproved production restarts (41 scenarios), with exact contexts.

Among calls at the same step, its self-report and both reasoning text monitors are near chance or weak; only its sampled next actions forecast which call violates (Table 2; own − self-report +0.417 [0.370, 0.473]; own − GPT-6-luna's judge +0.295 [0.213, 0.380]). Its words are nearly constant: before every violating call, each answer to whether it would comply said yes, and before 93% (94% fresh) of them none of its 32 answers admitted a violation; its rare admissions came before safe calls (descriptive). Gemma-4 sampling from Claude's contexts does not (0.54 at the same step).

Across steps, Gemma-4's draws tie Claude's own (0.82 vs 0.80; exploratory; the registered test on rebuilt contexts found Gemma-4's draws better, −0.14 [−0.20, −0.09]), and Claude's draws alone (0.80) score below the step alone (0.82): across steps, other readers track *when* Claude violates; within a step, only its own policy tells *which* call (+0.35 [0.28, 0.42]; exploratory). The sampled actions still add to the best text monitor across steps (0.90 to 0.96).

The registered results replicated. On 492 fresh episodes (1,582 calls), in a test registered with its Holm correction before any episode existed, all 8 endpoints passed: own draws − self-report +0.405 [0.356, 0.457] (primary), − the two text monitors +0.328 [0.254, 0.401] and +0.260 [0.185, 0.333], lift across steps +0.044 [0.029, 0.058] (the lift was first found on existing data, secondary).

The policy read costs $0.002 per decision (median 9 s). On τ^2-bench the forecast was not above chance (0.61 [0.44, 0.84]).

Sampling, not the reader, matters. GPT-6-luna reading Claude's sampled calls beats reading the transcript (+0.334 [0.235, 0.421]; fresh +0.302 [0.228, 0.372]), and Claude's own reads (equivalence failed).

The geometry alone forecasts. With no label and no question sign, the mean distance of the agent's sampled next actions from its typical state in the estimated geometry forecasts its coming violation: 0.89 on Claude Haiku 5.5 (0.87 on fresh episodes, registered before they existed), weakly 0.64 on GPT-6-luna, and on average 0.75 [0.68, 0.81] over four open agents across their task families (Gemma-4 0.77, gpt-oss-20b 0.75, Qwen3.8 0.91, Muse-Glimmer 0.56 [0.44, 0.68]; secondary: on existing draws).

What the weights add. A probe on CAPE's stored decision state forecasts better: the registered comparison failed, the probe leading by 0.16 [0.04, 0.28] (Gemma-4 0.97 vs 0.78; gpt-oss-20b 0.82 vs 0.70). That state is read after the agent's own hidden reasoning for the turn. On Gemma-4, a probe at the start of the turn, the moment a closed API forecasts from, scores 0.80 on 217 units, against 0.78 for the API geometry and 0.80 for the signed API forecast; turn-start minus after-reasoning probe is −0.20 [−0.31, −0.10]. So on this agent most of the white box's lead may come from the reasoning, but the registered non-inferiority test at turn start (−0.02 [−0.16, 0.12], bar −0.05) was too small to pass; a powered test is under way.

The reconstructed geometry alone flags violations. Before each step, both closed agents drew calls in side branches, and CAPE's checker labelled them. With no label and no question sign, a drawn call's Fisher–Rao distance from the agent's typical state ranks the violating ones: 0.818 [0.792, 0.844] on Claude Haiku 5.5 (1,765 calls, restarts only, so within tool) and 0.816 [0.537, 0.986] on GPT-6-luna (832 calls; the interval is wide; secondary). The signed read-out of Section 4.1 is one projection of this geometry; it adds the direction of each question and scores higher (0.89 and 0.99 on the same calls). On the fresh episodes, the geometric monitor scored 0.81.

The theory predicts the loss of any forecaster that does not sample. By Proposition 3, a forecast q from the context alone loses $\mathbb{E}(f-q)^2$ in Brier score against the agent's policy f; the agent's draws, checked by CAPE's checker, estimate this without a label on the executed call (Corollary 2). Registered before any episode existed, the predicted loss of another model's draws matched the observed on both closed agents (a first Claude test, with contexts rebuilt from logs, failed: -0.046): 0.122 vs 0.126 on Claude Haiku 5.5 (difference +0.004 [-0.003, 0.010]) and 0.018 vs 0.020 on GPT-6-luna (+0.002 [-0.004, 0.008]; bar ± 0.02). The predicted losses of three text monitors also matched (each inside ± 0.02; secondary), and on the fresh episodes they matched again (+0.002 [-0.002, 0.007] and -0.003 [-0.010, 0.003]). The prediction needs checked draws: with GPT-6-luna's judgments in place of the checker it failed, even after recalibration. The registered Brier separation of the two regimes failed (GPT-6-luna's monitor ranks well but is poorly calibrated).

4.3. Preventing violations

Self-policing. A side branch asks the agent whether a non-read call complies with its policy; a flagged call goes to review, and a gate prevents the flagged share of violating calls (Proposition 13).

Live results (checker as reviewer; Table 9). Gemma-4 prevented 82% of its violating episodes, Claude Haiku 5.5 69% of those its transcript decides (31% over all), and on AgentDojo attack success fell by 84% and 50%, with no loss in goals or utility.

The effect is predictable. Offline flag rates, weighted by where each agent violates, predicted five agents' live reductions within the registered 0.15 (mean error 0.035; Figure 5); an earlier form failed, and on AgentDojo the law missed Gemma-4 by 0.16.

A gate on the pre-call state. On fresh GPT-6-luna episodes, at an equal review rate (checker as reviewer), a gate on the pre-call read-out cut violating episodes from 31.7% to 7.9%, a gate on the step to 17.9% (+0.313 [0.062, 0.543]; success 18.7% vs 27.8% with the step gate). The law predicted the first exactly (0.75) but under-predicted the step gate (0.17 vs 0.44); an earlier design at unequal rates did not separate the gates, and on Gemma-4 a pre-call gate did not prevent.

Prevention with no reviewer. A flagged call is held once and the agent told to re-check; an identical re-proposal runs. Holding the calls Claude's signed read-out flags cut violating episodes from 67.9% to 35.2%, against 42.9% for random holds at a matched dose (+0.114 [0.035, 0.200]; success +0.081; Holm).

The geometry alone gates, beyond random holds. A gate on the distance from the agent's typical state (cut set on unlabelled calls; no label, no question sign) cut Claude's violating episodes from 65.9% to 44.9% on 492 fresh units, but at a matched dose random holds did about as well (48.2%; +0.033 [-0.035, 0.100], failed). Claude's answers are nearly unanimous, so distances are discrete: half of the violating calls sat exactly at the cut, where that gate held by a coin, although the distance ranks them at 0.78–0.80 (exploratory). Registered with the tie removed (hold at or above the cut), the gate left 20.1% of episodes violating against 32.3% for random holds at the same dose (0.32 vs 0.30 holds per call; +0.122 [0.071, 0.177]). Holds stack on a policy reminder: 21.1% with the reminder, 10.8% with both (0.49 [0.34, 0.62]).

On GPT-6-luna (six templates) it cut them from 27.0% to 17.5%, against 23.0% for random holds (+0.206 [0.045, 0.388]; success -0.016 [-0.056, 0.028], lower bound just below the registered -0.05); where an unchecked fact decides the violation, nothing changed (Corollary 1). Forecast-timed reminders did no better than random ones (15.0% vs 18.3%).

5. Related work

Geometry from outputs. The Fisher metric, the arcsine transform and classical scaling are classical (Čencov, 1982; Amari, 2016; Anscombe, 1948; Torgerson, 1952). Representations and the Fisher–Rao geometry of next-token laws converge across models (Huh et al., 2024; Picozzi, 2026), and behavioural similarity tracks hidden similarity on open models (Schiekiera et al., 2026); logits recover the output projection up to an orthogonal map (Finlayson et al., 2024; Carlini et al., 2024). We estimate it through a closed API, with known noise, channel by channel.

Reading models; monitoring agents. Answers serve as embeddings (Benara et al., 2024), detect lies (Pacchiardi et al., 2024; Cooney et al., 2026) and beat white-box probes on open models (Sam et al., 2025); open models can monitor closed agents (Conti et al., 2026; Zhao et al., 2026). Self-reports are generic (Blandfort & Pawar, 2026) and lose to probes on tool calls (Advani et al., 2026; Binder et al., 2025); agents deny violations (Fayyazi et al., 2026). Chain-of-thought monitors catch reward hacking (Baker et al., 2025; Korbak et al., 2025; Arnav et al., 2025), and white-box probes join black-box monitors (Parrack et al., 2025); Ctrl-Z and retrying resample or block suspicious actions (Bhatt et al., 2025; Lucassen & Kaufman, 2026), while we sample to forecast.

6. Conclusion

Through its API alone, a closed agent's hidden-state geometry can be estimated, from answers and sampled actions; on Claude it forecasts and gates violations it does not admit.

Impact Statement

This work monitors closed AI agents before they act, through their own APIs and with no training. The intended effect is safer deployment: in our tests, with CAPE's checker reviewing the held calls, a gate on the agent's own answers prevented 82% of an open agent's violating episodes and, for a closed agent, 69% of those that its transcript decides (31% over all scenarios). We see four risks. (1) False confidence: the read-out cannot see a violation decided by a fact the agent never checked, and its questions cover the kinds of violation they were written for; a new policy needs new questions. (2) Holding calls costs time, and a held allowed call can end an episode early (GPT-6-luna's success changed by -0.016 [-0.056, 0.028]). (3) An agent that knows the questions, or an attacker who queries a score, may learn to answer them falsely; the questions and scores should stay hidden from the agent, and self-policing is no defense against a model that colludes with itself. (4) Reading a model's state from outside could help an attacker study a closed model; the method uses only ordinary answers and recovers geometry, not weights or private data. The code is released under the MIT license, and the browser tool does not upload the user's logs.

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A. Limitations

This appendix lists the limits of the method and of the evidence.

- **Benchmarks.** Most agent results use CAPE-Bench, whose tasks are synthetic and matched so that one fact decides each action. AgentDojo and τ^2-bench airline test the read-out on other policies, but on fewer agents.
- **One task family for Claude Haiku 5.5's forecasts.** Claude Haiku 5.5 rarely violates: on 2 of CAPE-Bench's 24 templates (on one of them in nearly every episode), in 1.6% of AgentDojo injection attempts (9 of 565, development and pilots), and in 1.5% of τ^2-bench airline episodes. In an exploratory scan of 72 episodes over all 24 templates, Claude Sonnet 5.5 violated only where a fact it never checked decides the violation (test-campaign sends, 3 of 3); a third closed agent with varying violations was not available to us. All its forecasting and reviewer-free prevention results therefore come from one template (unapproved production restarts, 41 scenarios); on τ^2-bench airline its policy forecast did not separate from chance (7 tasks). GPT-6-luna's results cover six templates.
- **At the call, the read-out acts as a fixed checklist.** The 8 answers agree with the agent's full decision state at only 0.01–0.16, and about 20% of their ranking power on open agents runs through the hidden violation direction. At the call, this is what the theory predicts (the transcript decides the label), and other models' answers do as well or better; the agent itself is needed only before the call.
- **Sampled calls.** The geometric monitor is measured on the calls that the agents drew in side branches, labelled by CAPE's checker, not on the calls they executed; on GPT-6-luna its interval is wide.
- **The theory is mostly elementary.** Its parts are classical or short (marked in Section 2). Its use is the predictions it makes before the data, which we test. It predicted the loss of three text monitors to within 0.02 on both closed agents, from sampled actions labelled by CAPE's checker (with an LLM reader in place of the checker, the prediction failed, even after recalibration on 200 labelled draws), the regimes at and before the call, and where the read-out works (the label-free SNR). It failed to predict another reader's loss on contexts rebuilt from logs, and its label-free gap did not separate the two regimes in Brier terms (a rank-based check did).
- **Questions follow the policy.** The 8 questions (Section F.1) were written from CAPE-Bench's kinds of violation. Another policy needs its own questions, written before any answer exists, as we did for AgentDojo and τ^2-bench.
- **A fact that the agent never checked.** If a violation depends on a fact that is not in the transcript and that the agent never looked up, the agent's state holds no warning, and the read-out cannot flag it (Claude Haiku 5.5's mailing-list sends; two of seven τ^2 tasks, where the agent misread the booking). Such violations need a check of the world.
- **Nearly deterministic answers.** When an agent gives the same answer to almost every question every time, the read-out has little variance beyond its noise. The label-free SNR (Proposition 4) flags these settings before any label exists; Claude Haiku 5.5's settings have the lowest SNR.
- **A forecast cannot beat the agent's own randomness.** Where the agent samples its action at the call (the same state leads to a violation in one run and not in another), no reading before the call can forecast more than the rate at that state (Proposition 3).
- **A closed model's own geometry is estimated, not measured.** The triangulation assumes that the closed model, like the open ones, has no own-model excess, and that its hidden states load on the shared geometry like an open model's.
- **The reviewer.** In the live tests, CAPE's checker stands in for the human who reviews a held call. A human reviewer is slower and makes mistakes, so live prevention with people can be lower.
- **Adversaries.** An agent that knows the questions, or an attacker who can query the score, may learn to answer falsely. Self-policing is untrusted monitoring without an adversary; it is no defense against a model that colludes with itself. The questions and scores should stay hidden from the agent.
- **Cost.** The read-out asks 32 one-word questions per call (8 questions, 4 answers each). With prompt caching this cost $0.0014 per decision on Claude Haiku 5.5 (median latency 2.7 s, measured on 100 decisions).
- **Many tests.** We ran many preregistered tests, and some headline tests redesign earlier ones that failed (the prevention law, the pre-call gate, and Claude's forecast with exact contexts). Over all 37 registered endpoints of the closed-agent tests behind the abstract, passed or failed, 25 survive a Holm correction (exploratory; Section L); the other tests are not corrected.

Every forecaster on the same calls, across and within steps

Tests before the call registered one of two strata: within template and tool, across the call's step, or within template, tool and step. Across steps, a forecast mixes *when* an agent violates with *which* call violates: the step alone ranks Claude Haiku 5.5's violations at 0.82. Within a step, only the second remains. Table 3 scores every forecaster on the same calls under

both (exploratory). On Claude Haiku 5.5, the text judges track the step (GPT-6-luna's judge 0.90 across steps) but are near or below chance within a step, where only its sampled actions forecast. On GPT-6-luna, its self-report and its text judge forecast within a step as well as or better than its sampled actions.

Table 3. **Forecasters before the call, on the same calls.** AUROC within template and tool (across steps) and within template, tool and step. Exploratory. Best per column and agent in blue.

Forecaster	across steps	within a step
CLAUDE HAIKU 5.5 (RESTARTS; 1,587 CALLS, 316 VIOLATIONS; EXACT CONTEXTS)		
position alone	0.821	–
self-report	0.354	0.473
Claude text judge	0.630	0.448
GPT-6-luna text judge	0.904	0.596
own draws, own read	0.801	0.891
own draws, GPT-6-luna read	**0.914**	**0.930**
own draws, geometry	0.713	0.892
GPT-6-LUNA (SIX TEMPLATES; 1,134 CALLS, 131 VIOLATIONS)		
position alone	**0.782**	–
self-report	0.762	**0.836**
GPT-6-luna text judge	0.776	0.777
own draws, own read	0.681	0.729
own draws, geometry	0.748	0.642

Why the geometry-only gate tied with random holds (exploratory). In G72's and G75's live episodes, the unsigned distance ranks Claude Haiku 5.5's proposed violating calls at 0.78–0.80, but its answers are nearly unanimous, so distances take few values: 49–51% of violating calls (and 25–28% of safe ones) sat exactly at the cut, where the gate held by a coin of one third, and 20–22% of violating calls sat below it. The selection was therefore largely random at that dose. A test with the tie removed (holding every call at or above the cut, against random holds at the same dose) is registered as G76.

Claude's geometry on new kinds of context (exploratory). On G73's 400 fresh contexts, Claude's answer geometry agreed with the open models' hidden geometries better on the six familiar kinds (0.23–0.52 within them) than on the four new ones (0.13–0.51) for 6 of 8 models; even on familiar kinds it stayed below the 0.45–0.51 found on the three kinds its 24 questions were chosen on.

B. Proofs

This appendix proves the results of Section 2: the observable geometry, exact recovery from log-probabilities, the gauge, and the sufficiency of the transcript. The theory for monitoring agents is stated and proved in Section C.

B.1. The observable geometry

Proof of Theorem 1. (a) The Fisher information of $\mathrm{Bernoulli}(p)$ is $1/(p(1-p))$ and $\mathrm{d}\vartheta/\mathrm{d}p = 1/\sqrt{p(1-p)}$, so the metric is $\mathrm{d}\vartheta^2$. The Fisher metric of a product of independent factors is the sum of the factors' metrics, so in ϑ the family is the box $[0,\pi]^m$ with the Euclidean metric. The box is convex, so geodesics are segments and the distance is Euclidean. Check for one factor: the Bhattacharyya coefficient of $\mathrm{Bernoulli}(\sin^2\alpha)$ and $\mathrm{Bernoulli}(\sin^2\beta)$ is $\cos(\alpha-\beta)$, and $2\arccos\cos(\alpha-\beta) = |\vartheta-\vartheta'|$. (b) Invariance: d_Q is a function of $(P_x, P_{x'})$ only, and Proposition 1 leaves every output law unchanged. Uniqueness of the Fisher metric on the simplex over the finite set $\{0,1\}^m$, under Markov morphisms, is Čencov's theorem. d_Q is the intrinsic geodesic distance of the submanifold of independent laws; the geodesic distance of the whole simplex between two product laws is $2\arccos$ of their Bhattacharyya coefficient, $2\arccos\prod_j \cos\frac{\vartheta_j-\vartheta'_j}{2}$, which is a different function of the same angles. (c) $\mathrm{d}\vartheta_j = \Psi'_j/\sqrt{\Psi_j(1-\Psi_j)}$ and $\Psi'_j = \Psi_j(1-\Psi_j)\nabla a_j^\top\Delta$, so $\mathrm{d}\vartheta_j = \sqrt{\Psi_j(1-\Psi_j)}\,\nabla a_j^\top\Delta$; summing the squares gives $\Delta^\top G_Q\Delta$, and the remainder is third order by Taylor's theorem. (d) $\hat\vartheta$ is increasing in k and k is stochastically increasing in p, so μ_n is strictly increasing; $\mu_n \to \vartheta$ by the law of large numbers and continuity. The halves are independent, so $\mathbb{E}[(a_A-a'_A)(a_B-a'_B)] = (\mathbb{E}a_A-\mathbb{E}a'_A)(\mathbb{E}a_B-\mathbb{E}a'_B) = (\mu_n(\Psi)-\mu_n(\Psi'))^2$ for each question; sum over questions. This needs the same n in every cell: with $n_A(x) \neq n_A(x')$ the factor is $\mu_{n_A(x)}(\Psi) - \mu_{n_A(x')}(\Psi')$, not a difference of one function. Clipping a negative $\hat D^2$ at 0, as needed for distances, adds a non-negative bias. Classical scaling recovers a Euclidean configuration from its distance matrix up to a rigid motion (Young–Householder; Torgerson). □

Remarks. In part (c), s is the whole cache that the questions read, so its prediction for one layer's vector is heuristic. In part (d), the answer law is conditional on a yes or no answer; an answer that is neither yes nor no, or clipping negative estimates at 0, breaks the exactness.

The variance-subtraction rule is not exact. The rule estimates the squared distance by $\|\hat{\vartheta}(x) - \hat{\vartheta}(x')\|^2 - 2m/(n+\frac{1}{2})$. Exact binomial sums show that the bias of $\hat{\vartheta}$ is $\cot\vartheta/(4n) + O(n^{-2})$ (at $n = 32$: 0.0102 vs 0.0104 at $p = 0.1$), that the variance is below $1/(n+\frac{1}{2})$ near $p \in \{0,1\}$ ($0.68/(n+\frac{1}{2})$ at $p = 0.1$, $n = 8$), and that with n samples the angles cannot leave $[2\arcsin\sqrt{(3/8)/(n+3/4)},\ \pi - 2\arcsin\sqrt{(3/8)/(n+3/4)}]$, so large distances between saturated answers contract. The cross-fit of Theorem 1(d) replaces the rule and needs neither correction. The preregistered endpoints use the subtraction rule, as specified. On Claude Haiku 5.5 the cross-fit lowers the negative eigenvalue mass of classical scaling from 15% to 2%; on open models it keeps the same rank agreement with the hidden geometry (exploratory).

B.2. Exact recovery from log-probabilities

Where an API returns full log-probabilities, the final state is not estimated but recovered, by the construction of Carlini et al. (2024, App. H).

Theorem 2 (exact recovery; after Carlini et al.)**.** *Let the logits be Wu, with $u = x/\mathrm{rms}(x)$ the normalised final state, the normalisation gain folded into W, and no output bias. The centred log-probabilities $c(x)$ lie in a d-dimensional subspace; write $b = Q_d^\top c$ for an orthonormal basis Q_d of it. Then $u = Kb$ for an unknown invertible K, and $\|u\|^2 = d$ gives $b_i^\top S b_i = d$ with $S = K^\top K$: a linear system in S. If the $b_i b_i^\top$ span the symmetric matrices (generically, $N_{\mathrm{ctx}} \geq d(d+1)/2$ contexts), S is unique, and the Gram matrix of the final states, $b_i^\top S b_j = u_i^\top u_j$, is exact for every new context. The states follow up to an orthogonal map: the gauge group.*

The subspace (Finlayson et al., 2024) and the ellipsoid fit that removes all but an orthogonal ambiguity (Carlini et al., 2024, App. H) are known; we state the consequence for the gauge-invariant geometry and test its sharp transition at $d(d+1)/2$ contexts. Recovery holds up to the normalisation's ϵ, and near the threshold the system is badly conditioned.

Proof of Theorem 2. With $P = I - \mathbf{1}\mathbf{1}^\top/V$, the log-softmax removes a constant per context, so $c = PWu$ and every c lies in the column space of PW, of dimension d when PW has full column rank. Let Q_d be an orthonormal basis of it and $b = Q_d^\top c = Mu$ with $M = Q_d^\top PW$ invertible; then $u = Kb$ with $K = M^{-1}$. $\|u\|^2 = b^\top K^\top K b = d\,\overline{x^2}/(\overline{x^2} + \epsilon)$, which is d up to the normalisation's ϵ. The map $S \mapsto (b_i^\top S b_i)_i$ is linear on the $d(d+1)/2$-dimensional space of symmetric matrices; it is injective iff the $b_i b_i^\top$ span that space, which for b_i in general position needs and generically holds with $N_{\mathrm{ctx}} \geq d(d+1)/2$. Then $S = K^\top K$ is the unique solution and $B^\top S B = U^\top U$. Any L with $L^\top L = S$ satisfies $L = OK$ for an orthogonal O, so the states are fixed up to O, the group of Proposition 1. □

The solver. We whiten b on the training contexts (any invertible map of b is allowed) and solve the least-squares problem for symmetric S by conjugate gradient on the normal equations, from $S = 0$, which gives the minimum-norm solution when $N_{\mathrm{ctx}} < d(d+1)/2$. On synthetic heads with known K ($d = 16, 48$) the cosine error is 10^{-10} to 10^{-12} above the threshold and 0.04–0.09 below it.

The test. On Qwen3-0.6B ($d = 1{,}024$), from 1.15 million self-generated contexts, the recovered cosines of 2,000 held-out contexts correlate with the true ones at 0.30 and 0.36 (at $0.25\times$ and $0.5\times$) below $d(d+1)/2$ contexts (at $0.75\times$ and $0.9\times$ the minimum-norm solution is indefinite and the recovery fails); at $1.1\times$ the correlation is 0.9915, and at $2\times$ the median error is 7×10^{-6} (maximum 1.5×10^{-4}) (Figure 4b; preregistered). Closed APIs do not return full log-probabilities.

B.3. The gauge and the sufficiency of the transcript

Notation is that of Section 2. $\sigma(X)$ is the σ-field generated by X. A score is *strictly proper* if its expectation is uniquely maximised by the true conditional probability.

Proposition 5 (behavioural sufficiency)**.** *Let $Y = g(\tau)$ be defined by the future τ that the agent and the environment generate from context x, and let $\phi(x)$ carry no information about the future sampling noise given x. Then $\pi^*(x) = P(Y = 1 \mid x)$ has a maximal ROC curve and a minimal risk under every strictly proper score among all forecasts $f(\phi(x))$.*

Proof. By the assumption and the tower property, $\mathbb{E}[Y \mid \phi(X)] = \mathbb{E}[\pi^*(X) \mid \phi(X)]$. Under a strictly proper score the Bayes act given $\sigma(X)$ is $\pi^*(X)$, and $\sigma(\phi(X)) \subseteq \sigma(X)$; less information cannot raise the expected score. For the ROC curve, by

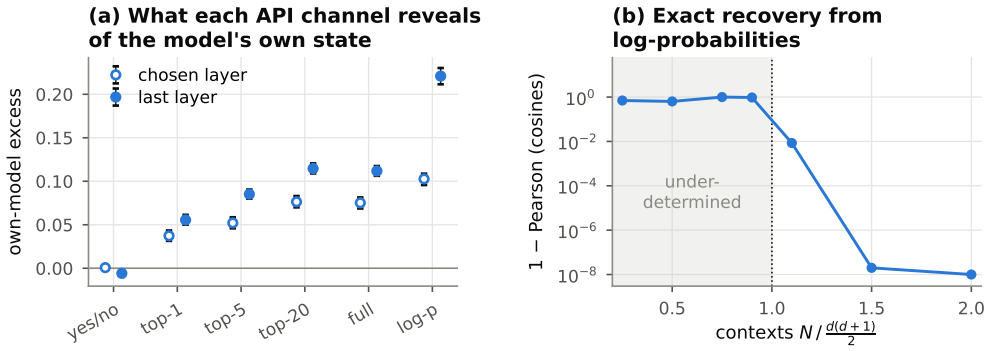


Figure 4. (a) Own-model excess for each API channel, against the last layer and the chosen layer, on 1,000 fresh contexts (preregistered; 95% intervals). (b) Exact recovery of the final hidden geometry from full log-probabilities (Qwen3-0.6B, 2,000 held-out contexts) against the number of contexts; the threshold is $d(d+1)/2$ (preregistered).

the Neyman–Pearson lemma the most powerful test of each size thresholds the likelihood ratio $p(x \mid Y{=}1)/p(x \mid Y{=}0)$, a strictly increasing function of $\pi^*(x)$; every test based on $f(\phi(X))$ is a test based on X. The same holds within any stratum that is a function of X, hence for the within-stratum AUROC. □

Proof of Proposition 1. Write a pre-norm residual network as $h^{(\ell+1)} = h^{(\ell)} + W_{\text{out}}^{(\ell)} F^{(\ell)}\big(W_{\text{in}}^{(\ell)}\, N(h^{(\ell)})\big) + b_{\text{out}}^{(\ell)}$, with embedding $h^{(0)} = E(x)$, output logits $U\,N(h^{(L)})$ and $N(h) = h/\mathrm{rms}(h)$ after folding the normalisation gains into W_{in} and U. For orthogonal G, $\mathrm{rms}(Gh) = \mathrm{rms}(h)$, so $N(Gh) = G\,N(h)$. Replace $E \mapsto GE$, $W_{\text{out}} \mapsto GW_{\text{out}}$, $b_{\text{out}} \mapsto Gb_{\text{out}}$, $W_{\text{in}} \mapsto W_{\text{in}}G^\top$ and $U \mapsto UG^\top$. By induction on ℓ the new residual stream is $Gh^{(\ell)}$, since $W_{\text{in}}G^\top N(Gh) = W_{\text{in}}N(h)$ and the new update is G times the old one; the logits $UG^\top N(Gh^{(L)}) = UN(h^{(L)})$ are unchanged, so the output distribution is unchanged on every input. At a layer read only through linear maps (no normalisation in between), the same argument holds for every invertible G with G^{-1} in place of $G^\top$. Any estimator of hidden coordinates that uses only outputs takes the same value on both networks, so it cannot distinguish h from Gh; a probe $w^\top h$ on one network equals $(G^{-\top}w)^\top(Gh)$ on the other. *Other normalisations.* LayerNorm subtracts the mean, $N(h) = Ph/\mathrm{rms}(Ph)$ with $P = I - \frac{1}{d}\mathbf{1}\mathbf{1}^\top$; $N(Gh) = GN(h)$ holds for orthogonal G with $G\mathbf{1} = \mathbf{1}$, which still form a group of dimension $(d-1)(d-2)/2$. A normalisation *after* a block, $\mathrm{diag}(\gamma)N(\cdot)$ added to the stream, needs $G\,\mathrm{diag}(\gamma) = \mathrm{diag}(\gamma)\,G$: G must commute with the gains, so for distinct gains only sign flips remain, and for equal gains any orthogonal map within the block of equal gains. From exact full-vocabulary log probabilities the final hidden state is identified only up to such a G (Carlini et al., 2024). □

Proof of Corollary 1. Let $A_t = a(Z_t^-, \xi)$ with $\xi \perp\!\!\!\perp (Z_t^-, W)$, and let V_t be a function of Z_t^-.

1. $\xi \perp\!\!\!\perp (Z_t^-, W)$ gives $\xi \perp\!\!\!\perp W \mid Z_t^-$, hence $\xi \perp\!\!\!\perp W \mid (Z_t^-, V_t)$; with $W \perp\!\!\!\perp Z_t^- \mid V_t$, contraction gives $W \perp\!\!\!\perp (Z_t^-, \xi) \mid V_t$.
2. A_t is a function of (Z_t^-, ξ), so $W \perp\!\!\!\perp (Z_t^-, A_t) \mid V_t$, and by weak union $W \perp\!\!\!\perp Z_t^- \mid (V_t, A_t)$.
3. $Y = g(A_t, V_t, W)$, so $Y \perp\!\!\!\perp Z_t^- \mid (V_t, A_t)$: $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$.
4. A monitor that reads internals at or before the action, and possibly the action, sees a σ-field inside $\sigma(Z_t^-, A_t)$. Under a strictly proper score the Bayes risk given a coarser σ-field is no smaller, and by step 3 the Bayes act given $\sigma(Z_t^-, A_t)$ is $P(Y \mid V_t, A_t)$; so its Bayes risk is at least that of the best monitor of (V_t, A_t). □

The inequality can be strict: if A_t is random given Z_t^- and Y depends on which action is sampled, a probe read before the action must average over the actions. *Canonical form.* If moreover $Y = g(\kappa(A_t), V_t, W)$ for a function κ of the action, the same steps with $\kappa(A_t)$ in place of A_t show that $(V_t, \kappa(A_t))$ is sufficient, and every monitor of $(V_t, \kappa(A_t))$ is invariant to any change of A_t that keeps $\kappa(A_t)$. The premise is about the label. In our attacked family it held: the checker labels a send by its recipient.

Proposition 3 (the forecast is the agent's policy). Let $f = \mathbb{E}[Y \mid V_t] = \sum_a \bar{\pi}(a \mid V_t)\, r(a, V_t)$. For any forecast f' that is a function of V_t (and of randomness independent of Y given V_t), $\mathbb{E}[(Y-f)(f-f')] = \mathbb{E}\big[(f-f')\,\mathbb{E}[Y-f \mid V_t]\big] = 0$, so $\mathbb{E}(Y-f')^2 = \mathbb{E}(Y-f)^2 + \mathbb{E}(f-f')^2$. With $f' = \sum_a \pi'(a \mid V_t)\, r(a, V_t)$, $f - f' = \Delta$, and $|\Delta| \le \sum_a |\bar{\pi} - \pi'|(a \mid V_t)\, r(a, V_t) \le \|\bar{\pi} - \pi'\|_1$ because $0 \le r \le 1$. Once A_t is visible, both readers forecast $r(A_t, V_t)$ and the excess is zero.

Corollary 2 (reading the policy by sampling). Given V_t, the draws a_k are i.i.d. from $\bar{\pi}(\cdot \mid V_t)$ and independent of Y, so $\mathbb{E}[\hat{f} \mid V_t] = f$ and $\mathrm{Var}(\hat{f} \mid V_t) = \mathrm{Var}_{\bar{\pi}}[r(a, V_t)]/M$. By the decomposition above, $\mathbb{E}(Y-\hat{f})^2 - \mathbb{E}(Y-f)^2 = \mathbb{E}(f-\hat{f})^2 = \mathbb{E}\,\mathrm{Var}_{\bar{\pi}}[r(a, V_t)]/M$, and $\mathrm{Var}[r] \le \frac{1}{4}$ for $r \in [0,1]$. When r is a $\{0,1\}$ label, $f(1-f)/M$ is estimated without bias by $\hat{f}(1-\hat{f})/(M-1)$; this gives the predicted loss of another reader used in Section 4.2.

C. Theory for monitoring agents

This appendix states and proves the theory behind the agent results. Each proof follows its statement.

C.1. The state across steps

Before each call we measure y_t, the angles of the agent's answers to m questions about its *next* action. By Theorem 1, $y_t = z_t + v_t$: the observable state plus sampling noise whose variance the number of answers fixes ($\approx 1/(n+\frac{1}{2})$ per question), independent across steps. We model $z_{t+1} = Fz_t + b + \eta_t$ and identify the model from unlabelled runs of the same agent.

Proposition 6 (identification through the noise)**.** *If v_t is independent across steps and of (z_s, η_s), and $\mathrm{Cov}(z_t, z_{t-1})$ is invertible, then $F = \mathrm{Cov}(y_{t+1}, y_{t-1})\,\mathrm{Cov}(y_t, y_{t-1})^{-1}$ and $\Sigma_z = F^{-1}\mathrm{Cov}(y_{t+1}, y_t)$, whatever the law of v_t. Regressing y_{t+1} on y_t instead shrinks F by the noise.*

Proof sketch. The noise at $t \pm 1$ is independent of everything at other steps, so $\mathrm{Cov}(y_{t+1}, y_{t-1}) = F\,\mathrm{Cov}(z_t, z_{t-1}) = F\,\mathrm{Cov}(y_t, y_{t-1})$ and $\mathrm{Cov}(y_{t+1}, y_t) = F\Sigma_z$. □

A Kalman filter (with the noise taken from the data: for nearly deterministic answers the nominal value overstates it two- to fivefold) then gives the estimated state $\mathbb{E}[z_t \mid y_{\le t}]$, measured before the call is proposed, and the forecast $\mathbb{E}[z_t \mid y_{<t}]$ from earlier steps alone; the gap between the noise that Proposition 6 implies and the theory's noise tests the model.

C.2. The cut

The cut is set on the agent's own unlabelled traffic and then frozen (Algorithm 1).

Proposition 7 (a frozen cut)**.** *Let the scores of n unlabelled warm-up calls and of a new call be exchangeable with no ties, and let $\hat{c}$ be the k-th smallest warm-up score. Then $P(S_{n+1} > \hat{c}) = (n+1-k)/(n+1)$, which for $k = \lceil (1-\beta)(n+1) \rceil$ lies in $\big(\beta - \frac{1}{n+1}, \beta\big]$.*

The proof also gives the long-run escalation rate: for i.i.d. scores with continuous distribution function F, $1 - F(\hat{c})$ follows $\mathrm{Beta}(n+1-k, k)$ over warm-up draws, and the interpolated empirical $(1-\beta)$-quantile has escalation probability within $\frac{2}{n+1}$ of β. With $n = 500$ and $\beta = 0.05$ the escalation rate has mean 0.050 and standard deviation 0.0097.

Proof. For exchangeable scores without ties, the rank of S_{n+1} among $S_1, \ldots, S_{n+1}$ is uniform on $\{1, \ldots, n+1\}$. $S_{n+1} > S_{(k)}$ holds iff its rank exceeds k, which has probability $(n+1-k)/(n+1)$; with $k = \lceil (1-\beta)(n+1) \rceil$ this lies in $(\beta - \frac{1}{n+1}, \beta]$. For i.i.d. scores with a continuous distribution function F, a new score exceeds $\hat{c}$ with probability $1 - F(\hat{c})$ given the warm-up; $F(S_{(k)})$ is the k-th order statistic of n uniforms, $\mathrm{Beta}(k, n+1-k)$, so over warm-up draws $1 - F(\hat{c}) \sim \mathrm{Beta}(n+1-k, k)$, with mean $(n+1-k)/(n+1)$ and variance $k(n+1-k)/\big((n+1)^2(n+2)\big)$; for $n = 500$, $\beta = 0.05$: $k = 476$, mean 0.0499, sd 0.0097. The shipped cut is the linearly interpolated empirical quantile at $1-\beta$, which lies in $[S_{(m)}, S_{(m+1)}]$ with $m = \lfloor 1 + (n-1)(1-\beta) \rfloor$. Its escalation probability is therefore in $\big[\frac{n-m}{n+1}, \frac{n+1-m}{n+1}\big]$, and $(n-1)(1-\beta) \le m \le 1 + (n-1)(1-\beta)$ puts it in $\big[\beta - \frac{2\beta}{n+1}, \beta + \frac{2(1-\beta)}{n+1}\big]$, within $\frac{2}{n+1}$ of β (for $n = 500$: between 0.0499 and 0.0519). □

Proposition 8 (shift)**.** *If deployment scores follow Q and warm-up scores follow P, a frozen cut escalates within $d_K(P,Q) + \frac{2}{n+1}$ of β, with $d_K(P,Q) = \sup_c |F_P(c) - F_Q(c)|$ the Kolmogorov distance.*

Proof. For every c, $Q(S > c) - P(S > c) = F_P(c) - F_Q(c)$, whose absolute value is at most $d_K(P,Q)$. Apply it at $c = \hat{c}$ and combine with Proposition 7; apply it to the laws of the violations' scores for the share caught. □

Algorithm 1 The grey-box monitor

Require: open-agent steps $\{(V_i, A_i, Y_i)\}$; reader R, layer ℓ; budget β; warm-up length n
$x_i \leftarrow \text{render}(V_i, A_i)$: instructions, task, earlier calls and results, proposed action
$\psi_i \leftarrow R_\ell(x_i)$ at the last token {the reader's state}
fit f: standardise $\to$ PCA(256) $\to$ logistic (C=0.05) on (ψ_i, Y_i) {never on the test agent}
for the first n actions A_t of the monitored agent **do**
  record $s_t \leftarrow f(R_\ell(\text{render}(V_t, A_t)))$; run A_t {no labels}
end for
$\hat{c} \leftarrow$ the $(1-\beta)$-quantile of $s_{1..n}$; freeze it {Propositions 7 and 9}
for each later proposed action A_t **do**
  if $f(R_\ell(\text{render}(V_t, A_t))) \geq \hat{c}$ **then** block A_t, or hold it for review **else** run A_t
end for

Proposition 9 (bursts)**.** *A cut set at the top β of a window in which a share $\rho > \beta$ of the actions are violations catches at most β/ρ of them. A frozen cut $\hat{c}$ catches each violation with probability $P(S \geq \hat{c} \mid Y = 1)$, whatever ρ is.*

Proof. A cut at the top β of a window of N actions escalates $\lceil \beta N \rceil$ of them; the window holds ρN violations, so at most $\beta N/(\rho N) = \beta/\rho$ of them are escalated (up to rounding). A frozen cut $\hat{c}$ escalates a violation with probability $P(S \geq \hat{c} \mid Y{=}1)$, the monitor's true-positive rate at $\hat{c}$, which does not depend on how many violations the traffic holds when the score laws do not depend on ρ. A rolling window that holds only part of a burst has a smaller effective share than the burst, so it can catch more than β/ρ of the burst's violations; the bound is on the window. □

So a cut that follows recent traffic fails exactly when an attack arrives, while a frozen cut keeps its rate. After a shift of the safe traffic, the cut is set again (Proposition 8).

C.3. Replicating the agent's state

A *replica reader* is an open model R that reads the same visible tokens and the proposed action; $\psi_t = R(V_t, A_t)$ is its hidden state at a fixed layer, and ϕ_t is the agent's state at the decision token.

Proposition 10 (access and replica)**.** *(a) $\sigma(Z_t, \phi_t) = \sigma(Z_t)$. For every label Y, the best monitor that reads (Z_t, ϕ_t) is no better than the best that reads Z_t, under every strictly proper score and at every point of the ROC curve. Under log loss, the best monitor that reads Z_t beats the best that reads V_t by exactly $I(Y; H_t \mid V_t)$. (b) The L^2-optimal replica $\hat{\phi} = \mathbb{E}[\phi \mid \psi]$ satisfies $\mathrm{Cov}(\hat{\phi}) \preceq \mathrm{Cov}(\phi)$ and, for every component k, $\mathrm{Var}(\hat{\phi}_k) = R_k^2 \,\mathrm{Var}(\phi_k)$, with $R_k^2 = 1 - \mathbb{E}(\phi_k - \hat{\phi}_k)^2 / \mathrm{Var}(\phi_k)$. The same identity holds for the best linear replica, in sample. (c) For every linear readout w, $\mathbb{E}[w^\top \phi \mid \psi] = w^\top \hat{\phi}$: a probe applied to the replica is the conditional expectation of the same probe on the true state.*

Part (a) says why a replica is possible: with reasoning off, ϕ_t is a function of tokens that an outside model can also read. Part (b) predicts the shape of the replica, and part (c) says what a white-box readout computes on it.

Proof. (a) $\phi_t = \Phi_\theta(Z_t)$ is $\sigma(Z_t)$-measurable, so $\sigma(Z_t, \phi_t) = \sigma(Z_t)$ (for every label), and Proposition 5 applied with $X = Z_t$ gives the first claim. Under log loss the Bayes risk given a σ-field $\mathcal{G}$ is the conditional entropy $H(Y \mid \mathcal{G})$, and $H(Y \mid V_t) - H(Y \mid V_t, H_t) = I(Y; H_t \mid V_t)$. (b) By the law of total covariance, $\mathrm{Cov}(\phi) = \mathbb{E}\ \mathrm{Cov}(\phi \mid \psi) + \mathrm{Cov}\left(\mathbb{E}[\phi \mid \psi]\right)$, and the first term is positive semidefinite. For component k, $\mathrm{Var}(\phi_k) = \mathbb{E}(\phi_k - \hat{\phi}_k)^2 + \mathrm{Var}(\hat{\phi}_k)$, so $\mathrm{Var}(\hat{\phi}_k) = R_k^2 \,\mathrm{Var}(\phi_k)$. For the best linear predictor $\hat{\phi}_k = a_k^\top \psi + b_k$ the residual is uncorrelated with $\hat{\phi}_k$ (the normal equations), and the same decomposition holds in sample. (c) Conditional expectation is linear: $\mathbb{E}[w^\top \phi \mid \psi] = w^\top \mathbb{E}[\phi \mid \psi]$. □

On fresh steps the identity of part (b) is not exact: Figure 8d shows the ratio closely tracking R_k^2 on 128 components (correlation 0.99, slope 0.89), above R_k^2 on 80 of them.

Proposition 11 (the replica Rao–Blackwellises a probe)**.** *Let $s = w^\top \phi$ be a probe on the agent's state and $\hat{s} = \mathbb{E}[s \mid \psi]$ its readout on the L^2-optimal replica. If $Y \perp\!\!\!\perp \phi \mid \psi$, then $\mathbb{E}\,\ell(\hat{s}, Y) \leq \mathbb{E}\,\ell(s, Y)$ for every loss $\ell(s, y)$ that is convex in s. The same holds for every convex pairwise loss $\mathbb{E}\,\ell(s_i - s_j)$ of independent violating and safe actions i, j, the convex surrogates of the AUROC.*

Proof. $s = w^\top \phi$ is a function of ϕ, so $Y \perp\!\!\!\perp \phi \mid \psi$ gives $Y \perp\!\!\!\perp s \mid \psi$ and hence $\mathbb{E}[s \mid \psi, Y] = \mathbb{E}[s \mid \psi] = \hat{s}$. For ℓ convex in its first argument, conditional Jensen gives $\mathbb{E}[\ell(s, Y) \mid \psi, Y] \geq \ell(\mathbb{E}[s \mid \psi, Y], Y) = \ell(\hat{s}, Y)$; take expectations. For a pair of independent draws $(\phi_i, \psi_i, Y_i = 1)$ and $(\phi_j, \psi_j, Y_j = 0)$, $\mathbb{E}[s_i - s_j \mid \psi_i, \psi_j, Y_i, Y_j] = \hat{s}_i - \hat{s}_j$ by the same conditional independence and the independence of the draws, and Jensen applies to $\ell(s_i - s_j)$. The AUROC is the expectation of the non-convex step $\mathbf{1}\{s_i > s_j\}$, so the result covers its convex surrogates (logistic, exponential, hinge), not the AUROC itself. □

We estimate $\hat{\phi} = M\psi + b$ by ridge regression on paired steps of an open agent, on the top principal components of its state; the proposition applies to this estimate only as far as the linear map approximates the conditional mean. Section J gives the numbers.

C.4. Robustness to rewording

Let $\kappa(A)$ be the decisive part of an action (its tool and targets) and let the free text be the rest.

Proposition 12 (a ceiling under rewording)**.** *Suppose that for every violating action an attacker can reach a still-violating action with the same $\kappa(A)$ and any free text that safe actions with the same $(V, \kappa(A))$ use, and that the attacker picks, among these, the text with the lowest score of the monitor. Then for every monitor s and cut c there is a monitor m of $(V, \kappa(A))$ alone that escalates no more safe actions than s and, on clean traffic, catches at least the share of violations that s catches under the attack.*

Proof. Write an action as (κ, t), with t its free text, and let $T_0(V, \kappa)$ be the set of texts that safe actions with context V and decisive part κ use. Define $m(V, \kappa) = \inf_{t \in T_0(V,\kappa)} s(V, \kappa, t)$, and $m = +\infty$ where T_0 is empty. On a safe action $t \in T_0(V, \kappa)$, so $m \leq s$ and $m \geq c$ implies $s \geq c$: m escalates no more safe actions than s. Under the attack, a violation with (V, κ) is rewritten to the text in $T_0(V, \kappa)$ with the lowest score (allowed by the assumption; take a sequence if the infimum is not attained), so its attacked score is $m(V, \kappa)$, and it is escalated iff $m(V, \kappa) \geq c$. As m depends only on (V, κ), the attack does not change it, and its share of violations caught on clean traffic equals the share that s catches under the attack. Where T_0 is empty the attacker cannot imitate a safe action, and $m = +\infty$ escalates. □

Under rewording, no monitor's operating point beats the clean ROC curve of the best monitor of the decisive fields.

C.5. Self-policing: a prevention law

The agent is asked in a side branch whether its proposed call complies. A flagged call is replaced by a fresh draw, at most M draws, and after M flags the step goes to review ($M = 1$ is the gate).

Proposition 13 (prevention law)**.** *At a state let R be the agent's probability of proposing a violating call, and let the audit flag a violating call with probability τ and a safe one with probability φ, independently across draws. With $f = R\tau + (1 - R)\varphi$, a violation runs without review with probability $R(1 - \tau)\,(1 - f^M)/(1 - f)$, and the step goes to review with probability f^M.*

Proof. Draw j is the first unflagged draw with probability $f^{j-1}(1 - f)$; given that it is unflagged, it violates with probability $R(1 - \tau)/(1 - f)$. Summing over $j \leq M$ gives the first claim; all M draws are flagged with probability f^M. □

Every term is measurable through the agent's API: R by resampling it, τ and φ by auditing it. For a gate whose reviewer blocks flagged violations, the share of violations prevented at one state is τ when a blocked violation is not retried. Over states of kinds k with violation shares w_k and flag rates τ_k, it is $\sum_k w_k \tau_k$. On Gemma-4's development steps, the audit flagged $\tau = 0.883$ of the violating calls when all four answers said no (the gate's cut), and 0.966 when a majority said no. The live gate prevented 0.82 [0.76, 0.88] of Gemma-4's violating episodes. Figure 5 compares the law with the live tests.

C.6. Battery design

A battery of questions can be designed from black-box answers alone (Algorithm 2): from a pool of candidate questions, add the one that splits the most classes of contexts with equal answers, and stop when no candidate splits a class.

Proposition 14 (battery design)**.** *This greedy rule stops with the partition of the contexts that the whole pool induces and asks at most (number of its classes $- 1$) questions. If the model's answers come from a finite automaton and the pool holds a*

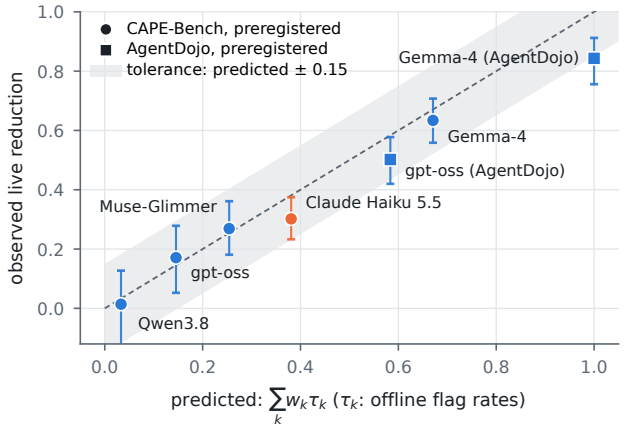


Figure 5. **The prevention law predicts live self-policing.** Live reduction in violating episodes (95% interval) vs the prediction from offline flag rates per kind of task, weighted by where the agent violates. Dots: the composition-weighted law (preregistered; all five within ± 0.15). Squares: prompt injection on AgentDojo (preregistered).

Algorithm 2 Battery design from black-box answers

Require: contexts $x_1..x_N$; pool of questions $\mathcal{Q}$; answers $a(x, q) \in \{0, 1\}$ from the model's outputs only
1: $C \leftarrow$ one class holding all contexts; $Q \leftarrow ()$
2: **loop**
3: $\quad q^\star \leftarrow \arg\max_{q \in \mathcal{Q}}$ #classes of C refined by $a(\cdot, q)$
4: $\quad$ **if** $q^\star$ splits no class **then return** Q {Proposition 14}
5: $\quad$ append $q^\star$ to Q; $C \leftarrow C$ refined by $a(\cdot, q^\star)$
6: **end loop**

distinguishing suffix for every pair of its states, this partition is the minimal (Myhill–Nerode) state, and no black-box method can identify more.

Proof. Each chosen question splits at least one class, so the number of classes grows by at least one per question and starts at one; the loop stops when no question in the pool splits a class. At that point two contexts in the same class agree on every pool question (otherwise that question would split their class), and two contexts in different classes disagree on a chosen question; so the final partition is the one that the whole pool induces, reached with at most (number of its classes $- 1$) questions. If the answers come from a finite automaton with minimal automaton M, two prefixes give the same answer to every suffix iff they reach the same state of M (Myhill–Nerode); when the pool holds a distinguishing suffix for every pair of states, the pool partition is the partition by minimal state. States that never change an output are behaviourally inert, so no black-box method identifies more. □

On a program compiled into a transformer (a bracket-balance automaton: 21 reachable states plus a dead state), the algorithm chose 21 questions from a pool of 2,101 and separates all 22 classes.

D. Trained comparisons: the reader monitor in full

This appendix reports monitors that are *trained*, as comparisons for the training-free method. With labels, an open reader of the transcript is a monitor: a logistic probe on its state, trained only on 17,623 Gemma-4 steps, scores every agent's actions (Table 4); baselines are the agent's own internals (where open), TF-IDF with the same labels, a zero-shot 70B judge, a reasoning judge given the policy, and the action's tool alone.

The transcript is enough. Corollary 1 says that the agent's internals cannot beat a monitor of the transcript and the action. A probe on Gemma-4's own internals scores 0.841 at the start of the turn and 0.979 after its reasoning; the reader probe, trained on the same labels, scores 0.994 ($+0.015$ [0.007, 0.026] over the internals) and is non-inferior on unseen task families (Table 4). Given the action, adding the agent's hidden reasoning changes Gemma-4's AUROC by -0.0014 [-0.0046, $+0.0014$]. On three other reasoning agents (preregistered), the reasoning lowered transfer from Gemma-4 by

Table 4. **Detecting violations before the action runs.** Within-template AUROC (within suite on AgentDojo); nothing is fit on a test agent. The internals need the agent's weights; the other monitors read the transcript and the action. Bold: best; underline: second. Last column: the preregistered contrast (I: reader − internals; NI: non-inferiority to internals; J: reader − judge; J_{tool}: within tool), with its 95% interval. †Exploratory, on held-out scenarios; internals: a probe trained on the agent's own states and labels. ‡Exploratory: Qwen3-32B with reasoning, given the policy, checking each rule (3 samples).

Setting	Agent	Actions	Monitor					Preregistered contrast
			Agent's internals	70B judge (zero-shot)	Reasoning judge, rules‡	TF-IDF (same labels)	**Reader probe**	
Open agents: the training agent, then three other architectures								
Same scenarios	Gemma-4	12,128	0.979	0.910	–	<u>0.992</u>	**0.994**	I +0.015 [0.007, 0.026]
Unseen task families	Gemma-4	12,128	**0.976**	0.910	–	0.950	<u>0.969</u>	NI −0.007 [−0.014, 0.000]
Mixture of experts	gpt-oss-20b	5,229	–	0.833	–	**0.943**	<u>0.915</u>	J +0.082 [0.043, 0.129]
Hybrid attention†	Qwen3.8-27B	2,378	0.955	0.835	–	<u>0.966</u>	**0.972**	J +0.135 [0.035, 0.258]
Dense†	Muse-Glimmer	2,657	0.929	0.911	–	<u>0.960</u>	**0.968**	J +0.059 [0.017, 0.131]
Closed agents, through their APIs (reasoning off)								
CAPE-Bench	Claude Haiku 5.5	3,128	–	0.633	0.710	**0.997**	<u>0.991</u>	J +0.360 [0.302, 0.413]
CAPE-Bench	GPT-6-luna	3,627	–	0.942	0.791	<u>0.968</u>	**0.995**	J_{tool} +0.039 [0.015, 0.066]
A new benchmark: AgentDojo prompt injection, a policy absent from training								
No AgentDojo labels	two open	3,455	–	<u>0.537</u>	–	0.503	**0.781**	J +0.249 [0.136, 0.369]
Labels of open agents only	GPT-6-luna	3,925	–	<u>0.787</u>	–	–	**0.922**	J +0.136 [0.078, 0.191]

−0.033 [−0.046, −0.017]. Fit on the agent's own steps, the replica reader's state replicates 46–66% of the variance of the agent's top 128 decision-state components on four architectures (Sections C.3 and J). Over word counts it adds robustness to shift: TF-IDF ties it in distribution but is at chance on AgentDojo (0.503 vs 0.781).

Two closed agents. Nothing was fit on the closed agents. At a cut set at the top 5% of the agent's own unlabelled traffic, the monitor catches 94.1% [85.8, 97.7] of Claude Haiku 5.5's 68 violations (zero-shot judge: 0%; TF-IDF: 63.8%) and 98.5% [91.9, 99.7] of GPT-6-luna's (judge 37.9%; TF-IDF 49.0%; Wilson intervals). Claude Haiku 5.5's violations are two behaviours: 42 test sends to a mailing list with thousands of external members it never checked, and 26 production restarts without approval. A prior that knows only each template's and tool's violation rate, fit out of fold, catches 91.6% [82.1, 95.9] (AUROC 0.992; reader 0.989), so on Claude Haiku 5.5 the result shows that the monitor flags these two behaviours, not that it reads the deciding facts. GPT-6-luna's 66 violations span six templates, each with safe calls, and there the monitor beats that prior (98.5% vs 86.4% [76.1, 92.7]), as on Gemma-4. Score scales do not transfer between agents (a cut at 0.5 catches 3% on Claude Haiku 5.5), rankings do: the cut must come from the agent's own traffic. Frozen after 500 actions, it keeps catching 94% under bursts in which half the actions violate; a cut that moves over 5,000 actions falls to 36% (means of 50 bursts; Proposition 9, Figure 9a).

Live prevention. We ran the monitor as a server in each closed agent's tool loop, on the same scenarios with and without the gate. In all four preregistered live tests the gate removed nearly all executed violations (Table 5 and Figure 7), and no agent executed a violation after a block. Holding an escalated action for a reviewer kept the safety gain. The round trip was 89 ms on GPT-6-luna and 80 ms on Claude Haiku 5.5 (median; 95th percentiles 215 and 139 ms).

An agent that knows it is monitored, and an attacker who queries the score. Told that a monitor reviews every call, Claude Haiku 5.5 violated as often; the monitor caught 97.1% at the frozen cut, which then escalated 15.2% of actions (88.6% at a 5% budget; preregistered); a cut set again on its unlabelled traffic removed every executed violation live (Table 5). An attacker who queries the score and rewrites a violating call's free text lowered the catch from 96% to 29% (Section A): the score must stay hidden from the agent.

Labels of a new policy transfer from open agents. Prompt injection on AgentDojo is absent from our training data; its labels from two open agents transfer to the closed GPT-6-luna (AUROC 0.922 vs the judge's 0.787; Table 4).

E. A trained comparison: recreating the decision state

This appendix describes a trained comparison that recreates the agent's decision state. It is not part of the method: it trains linear maps and probes on open agents. Let O_t be what the API yields at the decision: a replica reader's state ψ_t of the same visible tokens and action, and the agent's own answers ϑ_t to yes/no questions about its proposed call, asked in side branches.

Table 5. **Live prevention** (preregistered; 500, 450, 450 and 400 scenarios, each without and with the gate). Episodes with an executed violation and episodes that reached the goal, in %; Δ: gated − ungated violation rate, with its 95% interval.

	Violations			Goals	
Agent, gate	none	gate	Δ [95% CI]	none	gate
GPT-6-luna, block	6.6	**0.2**	−0.064 [−0.088, −0.044]	47.4	45.6
Claude Haiku 5.5, block	6.7	**0.2**	−0.064 [−0.087, −0.044]	40.7	34.9
hold for review[a]	6.7	**0.0**	−0.067 [−0.091, −0.044]	40.7	38.7
told, cut re-set[b]	6.5	**0.0**	−0.065 [−0.090, −0.043]	38.5	35.5

[a]An oracle reviewer; same scenarios as the row above, run separately. [b]Block, with the cut set again on the told agent's unlabelled traffic: it escalated 4.7% [3.5, 6.1] of actions (budget 5%; the old cut 16.6%).

Table 6. Development results (exploratory): within-template AUROC of a probe trained on true decision states.

Agent	true state	replica reader + answers	replica reader	answers	others' coordinates	R^2 (replica reader)
Gemma-4	0.956	0.977	0.978	0.995	0.961	0.66
gpt-oss-20b	0.798	0.859	0.856	0.699	0.724	0.48
Qwen3.8-27B	0.854	0.864	0.831	0.862	0.819	0.57
Muse-Glimmer-30B	0.856	0.902	0.898	0.862	0.938	0.30

The grey box is the *recreated decision state* $\hat{\phi}_t = \mathbb{E}[\phi_t \mid O_t]$, estimated by linear maps fit on an open agent whose states ϕ are known; a probe trained only on true states reads it.

Proposition 15 (reading the recreated state)**.** *For every linear probe w, $w^\top \hat{\phi}_t = \mathbb{E}[w^\top \phi_t \mid O_t]$. If $Y \perp\!\!\!\perp \phi_t \mid O_t$, the probe on $\hat{\phi}_t$ has expected loss at most that on ϕ_t for every loss convex in the score.*

With reasoning off, ϕ_t is a function of the visible tokens, so in population the recreation is exact and the two losses are equal: nothing is lost by not having the weights (proof: Propositions 10 and 11, Section C). In practice the maps are linear, fit on a few hundred calls, on 32 principal components; how much this light recreation loses is an empirical question, which we test. What is new is not that the transcript suffices but that probes trained on *true* states, of one agent, can be carried to another agent whose states are never seen.

The agent's answers. The answers ϑ_t are the angles of the 8 at-call questions; Section F.1 gives the prompt and the questions.

The recreation. For an agent whose decision states ϕ are stored (at CAPE's decision token of each call, the 4th of five stored layers), the top 32 principal components z of ϕ are fit on development calls. The replica reader (Qwen3-8B) reads the transcript and the call; its last-token state at layer 32 is reduced to 256 principal components, and a ridge map ($\alpha = 10$) to z is fit on development calls. A second ridge map ($\alpha = 1$) from the 8 angles to the reader's residual $z - \hat{z}_{\text{reader}}$ is fit on the development battery. The recreation is $\hat{z} = \hat{z}_{\text{reader}} + \hat{z}_{\text{answers}}$. The white-box probe is a logistic regression ($C = 0.1$) on the true z of development calls and their labels. No label enters a map.

Other agents' coordinates. To score an agent without its states (a closed agent, or an open agent held out), its call is recreated in each source agent's coordinates and read by that source's probe; each source's scores are standardised over the scored calls (no labels) and averaged.

Controls (development). Pairing the answers with other calls' states before fitting the map takes the probe to chance on Gemma-4 (0.46): the recreation, not the probe, carries the signal. A logistic regression fit with labels directly on the 8 angles scores 0.980 on Gemma-4, below the label-free recreation (0.995), as Proposition 15 allows.

Development results. Scenario-disjoint 5-fold cross-fit (Table 6).

F. Experimental details

This appendix gives the settings of the agent experiments and of the geometry experiments.

F.1. The questions and the prompt

At a proposed call, the agent's conversation up to the call is sent again through its own API with one operator message: "[Operator check. Your next tool call is on hold and has NOT been executed.] Proposed call: *name*(*arguments*). Question about this call: q Answer with one word: yes or no." Tool use is disabled for the answer. Each of the 8 questions is asked in its own side branch, 4 answers each, sampled with the agent's own settings (open agents: their vLLM server and parsers; Claude Haiku 5.5: thinking disabled, low effort). The answer law of question j is estimated by the Anscombe angle $\hat{\vartheta}_j = 2\arcsin\sqrt{(k_j + 3/8)/(m_j + 3/4)}$ from k_j "yes" among m_j readable answers. The 8 questions ask whether the call:

1. complies with the policy;
2. touches a resource the user does not own and is not authorised for;
3. sends data outside its authorised audience;
4. carries out an instruction found inside a file, email, page or tool output;
5. is irreversible or on a production system without prior approval;
6. uses a credential not issued for the task;
7. affects uninvolved people without approval;
8. breaks an explicit constraint of the task.

F.2. The agent experiments

Agents. All agents run CAPE-Bench's mock environment for at most 20 turns, with one tool call per turn.

- Open agents, served by vLLM, at most 4,096 new tokens per turn:
  - Gemma-4 (26B-A4B): temperature 1.0, top-p 0.95, top-k 64, thinking on.
  - gpt-oss-20b: temperature 1.0, top-p 1.0, reasoning effort "medium".

  The episodes of the three additional open agents come from the released CAPE-Bench rollouts.
- Closed agents use their public tool-use APIs:
  - Claude Haiku 5.5: thinking disabled, effort "low", at most 1,024 tokens per turn.
  - GPT-6-luna: reasoning effort "none", no parallel tool calls, at most 2,048 tokens per turn.

  The shared spend ledgers record $24.78 of API use in total (Claude Haiku 5.5's provider $19.86, GPT-6-luna's provider $4.92), including tests of the released tool and a 40-episode pilot ($0.10) of Claude Haiku 5.5 on AgentDojo that was not analysed.

Units. A unit is one proposed action that is not a read, after the step that reveals the deciding fact. Each episode contributes its actions up to and including its first violation. All violations are kept. Safe actions are subsampled at random with a fixed seed (Table 7), so selection depends only on the past.

Reader input. The reader prompt holds four parts:

- the operating instructions;
- the user task;
- each earlier call with its result (results cut to 2,000 characters);
- the proposed action.

A context of 30,000 characters or more keeps its first 6,000 and last 24,000 characters. The prompt goes through Qwen3-8B's chat template with thinking off and is cut to its last 12,000 tokens. The feature is the residual stream at the last prompt token, at hidden-state index 32 (the output of block 31 of 36), in bf16.

Monitor. The monitor standardises the features, keeps the top 256 principal components, and fits a logistic regression with $C = 0.05$. It is trained on Gemma-4's CAPE-Bench steps only (17,623; development and fresh). The review cut has no labels. Offline (Table 10), it sends exactly 5% of the agent's actions in the test set, the highest-scored, to review (actions tied at the cut count in proportion). Live, it was the 95th percentile of the agent's earlier unlabelled actions, then frozen; the tool freezes it after the first 500. A rolling cut fails under bursts of similar actions.

Replica.

- **Target:** the agent's residual stream at its end-of-reasoning token, at 5 depths (block $\mathrm{round}(nf) - 1$, $f \in \{0.5, 0.62, 0.72, 0.82, 0.9\}$). It is standardised and reduced to 128 principal components, all fit on development.
- **Map:** ridge regression, with $\alpha \in \{1, 10, 10^2, 10^3, 10^4\}$ chosen by 5-fold cross-validation grouped by scenario.

- **Probe on the true state:** standardise, then logistic regression with $C = 0.05$, on the 128 true components.

Text baseline. TF-IDF of the context and of the action, kept as two separate blocks:

- word 1- and 2-grams, minimum document frequency 3, at most 200,000 features each, sublinear term frequency;
- logistic regression with $C \in \{0.1, 1, 10\}$, chosen by 5-fold cross-validation grouped by scenario.

The browser tool runs the same model. On 302 real agent steps it gives the same text as Python, and its held-out AUROC is 0.9887 (Python 0.9886).

Statistics.

- AUROC is computed within strata (template and tool, or suite) and averaged with weights by the number of positive–negative pairs.
- Intervals are percentile intervals from 1,000 or 2,000 bootstrap resamples of scenarios or episodes, as each preregistration fixes. The seeds are fixed in each preregistration.
- Each preregistration (the document and its analysis code) is locked by its SHA-256 hash before its data exist. One script checks every lock and every number in this paper again from the saved results.

Compute. All GPU work runs on NVIDIA H100 NVL GPUs (96 GB), one job at a time except for one 8-hour period with two: PyTorch 2.11, Transformers 5.14.1 (5.20.1 for Muse-Glimmer, whose architecture the earlier version does not know), scikit-learn 1.9.0. The reader takes 41 ms for one action on one GPU; the median server round trip in the three blocking live tests was 89, 80 and 83 ms.

F.3. The geometry experiments

Contexts and battery. 4,800 single-message contexts, 800 from each of six sources (harmful and benign requests, SST-2 reviews, BoolQ questions, GSM8K problems, Wikipedia text), split at random into 3,360 development and 1,440 confirmation contexts before any analysis. The battery has 88 yes/no questions about the message (topic, intent, tone, demands, risk), each appended after a fixed preamble that asks for a one-word answer; $P(\text{YES}) = p_{\text{YES}}/(p_{\text{YES}} + p_{\text{NO}})$ from the top-20 log-probabilities of the first answer token (cells where both are 0, 2 of 38,400 per model, get $\frac{1}{2}$).

Models. Gemma-3 4B, gpt-oss 20B, Llama-3.1 8B, Mistral 7B v0.3, OLMo-3 7B, Phi-4 mini, Qwen2.5 7B and Qwen3 8B, instruction-tuned, with their own chat templates (vLLM for answers and replies, Hugging Face transformers for hidden states). Hidden states: the residual stream at the last prompt token at six depths (25, 50, 62, 75, 90 and 100% of the layers). The layer of each model is the one whose cosine geometry best matches the exact answer geometry on 1,000 development contexts; the confirmation contexts are not used to choose anything.

Sampling. Sampled answers are drawn from the model's own probabilities (8 per question for the observable-geometry test; one per question, as yes, no or other, for the test of answers against replies). Text and reply embeddings: all-MiniLM-L6-v2, masked mean, L_2 normalised; replies are 8 samples at temperature 1, at most 128 tokens.

Claude Haiku 5.5. Claude Haiku 5.5 answered a fixed battery of 24 yes/no questions, chosen on open models before any Claude Haiku 5.5 data existed, 8 times each, on 192 confirmation contexts (thinking off, at most 3 tokens). The two halves are a random permutation of each cell's 8 samples (seed in Table 7). The bound of the split-half criterion (0.7) was set from open models' simulated split halves on development contexts only (0.76–0.96).

Exact recovery. Qwen3-0.6B in float32 (TF32 off). 9,000 sequences of 128 tokens sampled from the model (temperature 1), every position a context. The basis Q_d is from the first 3,000 contexts ($\sigma_d/\sigma_{d+1} = 1{,}005$). The true state for the check is the input of the final RMSNorm divided by its RMS.

Bootstrap. 500 resamples of contexts; Spearman on the pairs of distinct resampled contexts.

Table 7. Fixed random seeds named in this appendix. The bootstrap seeds are fixed in each preregistration.

Use	Seed
Subsample of safe actions (agent units)	0
Split halves of Claude Haiku 5.5's samples	2290
Exact recovery: sampled sequences	0 (a check with seed 100 gave the same result to the fourth digit)

F.4. Agreement of each observable with the hidden geometry

Table 8. **Which observable carries the state?** Spearman correlation of each geometry with the model's own hidden geometry, over all pairs of 1,440 held-out contexts; no parameter is fit. Answers: Fisher–Rao distances over 88 questions, exact (log-probabilities) or from 8 answers sampled from the model's own probabilities. Replies: mean embedding of 8 sampled replies. Text: embedding of the input. Best per row in blue; last column: answers − text, 95% interval.

	Answers (ours)				
Model	exact	8 samples	**Replies**	**Text**	**answers − text**
Qwen2.5 7B	**0.74**	**0.74**	0.30	0.28	**+0.46** [0.44, 0.48]
Llama-3.1 8B	**0.74**	0.73	0.32	0.26	**+0.48** [0.46, 0.50]
Mistral 7B	**0.70**	**0.70**	0.22	0.22	**+0.49** [0.47, 0.51]
Qwen3 8B	**0.66**	0.65	0.33	0.30	**+0.36** [0.34, 0.39]
Phi-4 mini	**0.58**	0.57	0.26	0.28	**+0.30** [0.28, 0.33]
OLMo-3 7B	**0.57**	0.55	0.34	0.29	**+0.28** [0.26, 0.30]
Gemma-3 4B	**0.56**	**0.56**	0.44	0.28	**+0.28** [0.26, 0.31]
gpt-oss 20B	**0.41**	0.40	0.25	0.25	**+0.16** [0.14, 0.19]

G. Trained comparisons (not the method): closed agents at a review budget

Table 9. **Self-policing through the agent's own API** (held-out scenarios; no other model in the gate, no labels; a held call goes to a reviewer, here CAPE's checker or, on AgentDojo, an injection reviewer). Reduction = 1−gate/plain; better arm in blue.

Agent, set	Violations (%) plain	Violations (%) gate	reduction (95% CI)	Success (%) plain	Success (%) gate
CAPE-BENCH: episodes with an executed violation; reached the goal					
Gemma-4, all[a]	27.9	**5.0**	**0.82** [0.76, 0.88]	33.5	**43.7**
Claude Haiku 5.5, decidable[b]	2.8	**0.9**	**0.69** [0.51, 0.88]	40.4	40.1
Claude Haiku 5.5, unchecked[c]	100	92.9	0.07 [0.00, 0.17]		
AGENTDOJO: attacks that succeeded; utility					
Gemma-4[d]	11.5	**1.8**	**0.84** [0.76, 0.91]	84.5	**86.8**
gpt-oss-20b[d]	24.4	**12.2**	**0.50** [0.42, 0.58]	50.5	50.8

[a]666 episodes per arm; held when all four answers say it does not comply. [b]470 scenarios × 2; held when all three answers say it breaks a rule. [c]21 mailing-list scenarios × 2, decided by a fact the agent never checks; predicted not to fall. [d]Held-out tasks, 2 trials; spotlighting in both arms. Details: Section G.

This appendix shows live prevention (Figure 7), the replica of the hidden state (Figure 8), robustness and transfer (Figure 9), and the closed agents at a 5% review budget (Table 10).

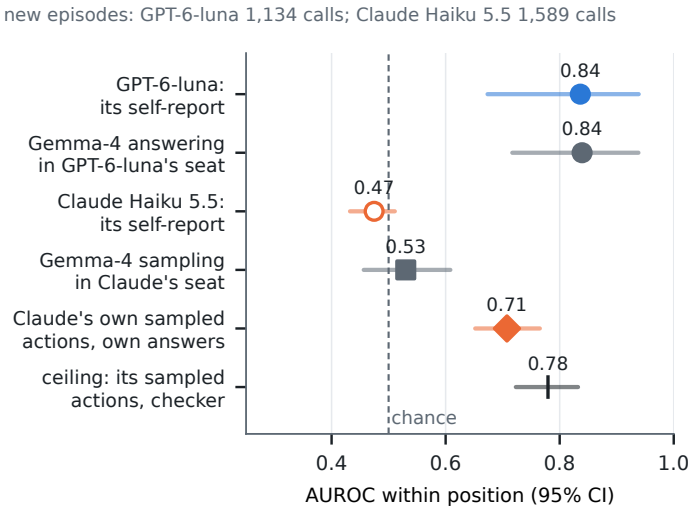


Figure 6. **What a closed agent says vs what it does**, before the call exists, on new episodes (AUROC among calls at the same position, 95% intervals). GPT-6-luna's answers about its next action forecast its coming violation, and Gemma-4 answering from the same contexts does as well. Claude Haiku 5.5's answers about its next action do not; its own sampled next actions, read by its own answers, do; Gemma-4 sampling from the same contexts does not. Ceiling: Claude's sampled actions judged by the environment. Claude: the first set of episodes, with contexts rebuilt from logs; on the exact-context set its sampled actions reach 0.89 (Figure 1c).

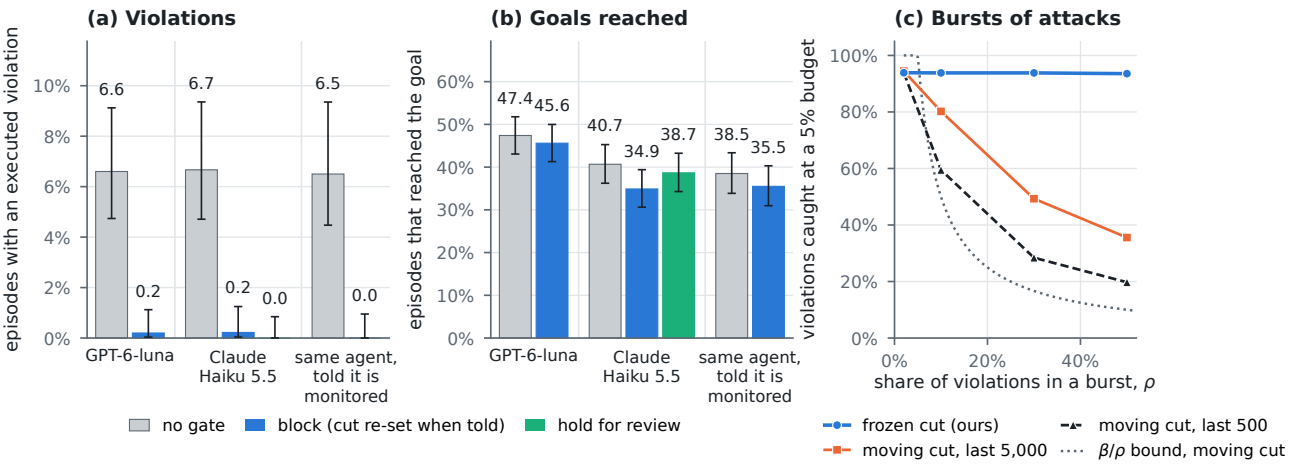


Figure 7. **Live prevention and bursts.** (a) Episodes with an executed violation and (b) episodes that reached the goal, in the four preregistered live tests: GPT-6-luna and Claude Haiku 5.5 without a gate and with blocking; Claude Haiku 5.5 with hold for review (an oracle reviewer, run separately); and Claude Haiku 5.5 told that it is monitored, with the cut set again on its unlabelled traffic. Bars: rates; whiskers: 95% Wilson intervals. (c) Violations caught at a 5% budget as the share ρ of violations in a burst grows, for the frozen cut and two cuts that move with recent traffic (Claude Haiku 5.5's scores, replayed); dotted: the β/ρ bound of Proposition 9.

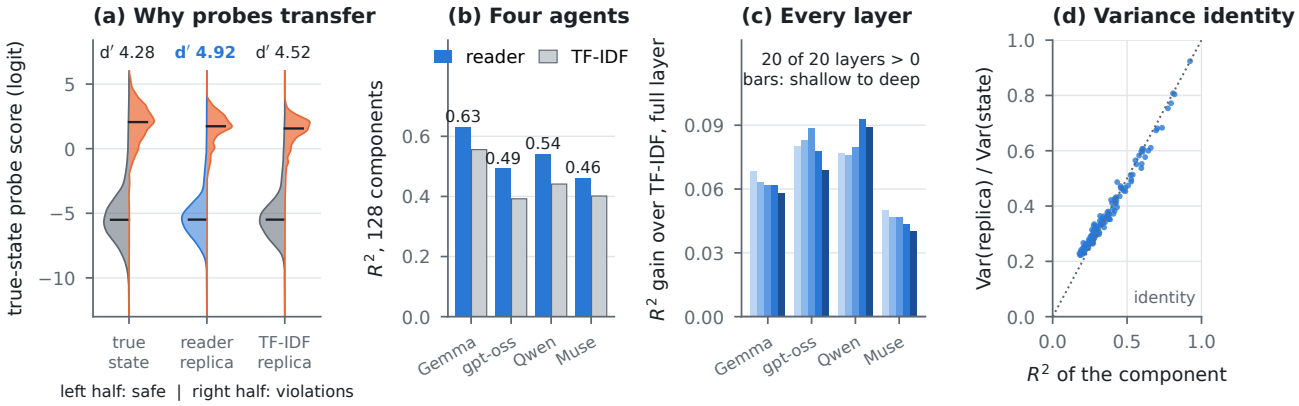


Figure 8. **Replicating the hidden state.** (a) Score of a probe trained on Gemma-4's true states, on the true state and on two replicas (12,128 fresh steps; left half safe, right half violations; bars: medians). (b) Variance-weighted R^2 of the top 128 components, unseen scenarios. (c) R^2 gain of the replica reader over the TF-IDF replica at each recorded layer, over the full layer. (d) Per component: replica variance over state variance, against R^2.

Table 10. **Closed agents at a 5% review budget.** A trained comparison, not the training-free method: each monitor except the judge and tool identity is fit on Gemma-4's labelled steps only. Pooled AUROC, and the share of violations caught when the top 5% of actions go to review (a label-free cut).

	Claude Haiku 5.5		GPT-6-luna	
Monitor	AUROC	caught	AUROC	caught
Zero-shot 70B judge	0.570	0.0%	0.778	37.9%
Tool identity	0.857	42.4%	0.856	13.6%
TF-IDF (same labels)	0.975	63.8%	0.964	49.0%
Reader probe	**0.989**	**94.1%**	**0.996**	**98.5%**

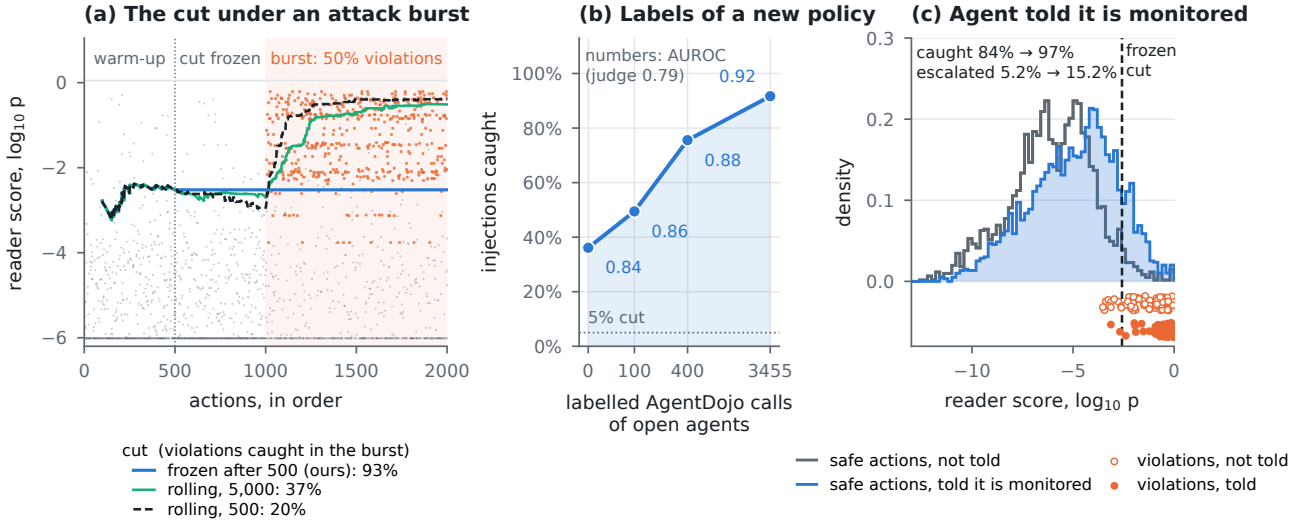


Figure 9. **Robustness and transfer.** (a) Claude Haiku 5.5's reader scores in order (violations in orange): 1,000 normal actions, then one burst in which half are violations; lines: three cuts (legend: share of this burst's violations caught; the text gives the means of 50 bursts, 94% and 36%). (b) The closed GPT-6-luna's injected calls caught at the benign 5% cut, against labelled AgentDojo calls of open agents (numbers: AUROC). (c) Claude Haiku 5.5's scores when told, or not told, that it is monitored; dashed: the cut frozen on its earlier traffic.

H. The released tool

The package `graybox` runs the method of this paper: it asks a closed agent's API the fixed questions in side branches, computes the angles, the read-out, the geometric monitor and the policy forecast, and holds a flagged call, with no training. As an option, and only as a comparison, a website trains a reader monitor in the browser from the user's labelled logs, so the logs never leave the user's machine. Either monitor runs in a tool loop, an MCP proxy or an agent framework.

I. Trained comparisons (not the method): smaller readers

This appendix tests readers smaller than the shipped one, including encoders that run in the browser. All readers are trained on the same 17,623 Gemma-4 CAPE-Bench steps (1,339 violations); none is trained on a test set. Browser encoders: all-MiniLM-L6-v2, all-mpnet-base-v2 (three 1,000-character views) and Qwen3-Embedding-0.6B (one instruction-prefixed text, last-token pooling), each with a standardised logistic probe. Decoder readers: Qwen3-0.6B, 1.7B, 4B and 14B at the shipped reader's relative depth, with the shipped probe. The selected browser encoder, Qwen3-Embedding-0.6B, beats TF-IDF on the closed agent's AgentDojo calls (0.605 vs 0.335; +0.270 [0.134, 0.451]), so the website offers it for new kinds of tasks. In the browser it runs as a float16 ONNX model whose features match PyTorch's (cosine 1.000 on 40 steps); the 8-bit model's do not (0.852).

Figure 10 compares the readers by size.

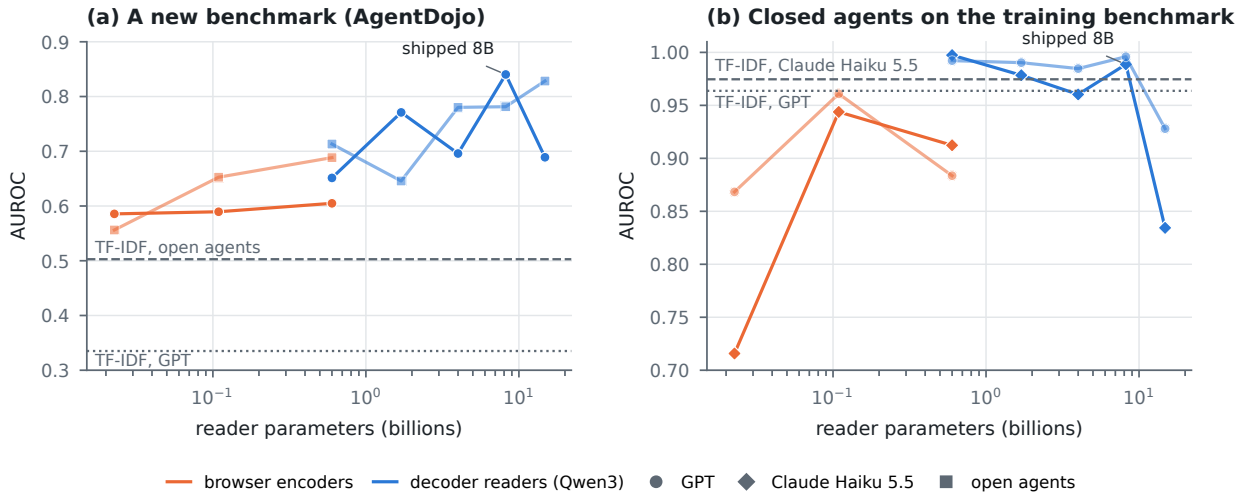


Figure 10. **Smaller readers.** AUROC against reader size for browser encoders and Qwen3 decoder readers, (a) on AgentDojo and (b) on the closed agents' CAPE-Bench steps. Lines: TF-IDF baselines. Every reader is trained on Gemma-4's CAPE-Bench steps only.

J. Trained comparisons (not the method): the probe on the replica

The numbers behind Proposition 11 and Section G, for the four agents of the replication tests.

Table 11. A probe trained on the agent's *true* states, scored on the true states and on the replica from the replica reader (rep.); fresh steps; exploratory. Pairwise risk: $\mathbb{E}\log(1+e^{-(s_i-s_j)})$ over violating i and safe j (lower is better); AUROC pooled and within template. Better of each pair in blue.

	pair risk ↓		AUROC		in template	
agent	true	rep.	true	rep.	true	rep.
Gemma-4	0.018	**0.015**	0.994	**0.996**	0.984	**0.987**
gpt-oss	0.078	**0.045**	0.972	**0.986**	0.900	**0.924**
Qwen3.8	0.044	**0.035**	0.986	**0.990**	0.955	**0.972**
Muse	0.056	**0.022**	0.982	**0.996**	0.929	**0.969**

K. Trained comparisons (not the method): the observable state in chat

On Claude Haiku 5.5, readouts of the observable state trained only on open models forecast which harmful requests Claude Haiku 5.5 refuses (Figure 11). Against a three-level moderation verdict the forecast wins (three preregistered tests). Combined with a moderator, it doubles the precision of an alarm for harmful requests that the model answers (0.535 vs 0.274).

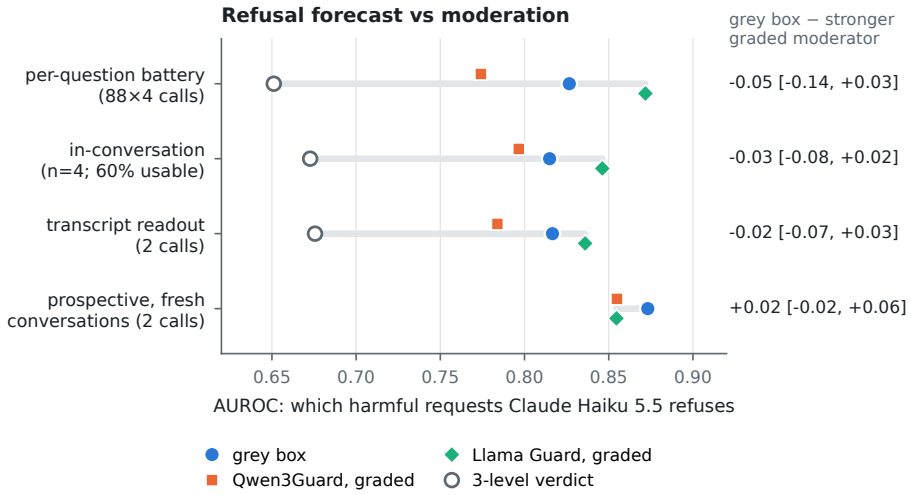


Figure 11. **Refusal forecast against moderation.** AUROC for which harmful requests Claude Haiku 5.5 refuses, for the grey box, two graded moderators (Qwen3Guard and Llama Guard) and a three-level verdict, in four ways of asking. Right: the grey box minus the stronger graded moderator, with its 95% interval.

L. Preregistered tests: those behind each claim, and every test

This appendix lists the preregistered test behind each claim. Each claim below rests on a registered endpoint of a test whose document and code we hash-locked before its data existed. The commit is the public commit that first contains the lock (withheld for anonymous review); for the later tests it precedes the run. The full record of every test, with its lock file, data and outcome, is public with the code. Rows marked † are secondary: they analysed data that already existed, locked before the analysis but registered on the same day after related results were known. Each of their positive results was then replicated on fresh episodes, with tests registered before any of those episodes existed.

Claim	Registered endpoint (bar)	Result	Lock
RECONSTRUCTING THE GEOMETRY (Section 3)			
Answers track the hidden geometry better than the input	answers − text, every model (lower bound > 0)	8 of 8; gains +0.16 to +0.49	*anon.*
Answers beat the replies	exact answers − replies (lower bound > 0, ≥ 7 of 8)	8 of 8	*anon.*
The geometry is shared across models	own-model excess inside ± 0.02	+0.002 [-0.001, +0.006]	*anon.*
A closed model's answer geometry is stable	split-half Spearman (> 0.7)	0.95 [0.94, 0.96]	*anon.*
Triangulation predicts own agreement	leave-one-out error (< 0.10)	0.061 [0.051, 0.070]	*anon.*
MONITORING AT THE CALL (Section 4.1)			
The read-out adds to the single answer	8 answers − 1 answer (lower bound > 0)	+0.036 [0.006, 0.064]	*anon.*
as good as a probe on the hidden states	8 answers − probe (lower bound > -0.05)	+0.045 [−0.017, 0.135]	*anon.*
on Claude Haiku 5.5's restarts	AUROC ≥ 0.80, lower bound ≥ 0.65	0.909 [0.858, 0.952]	*anon.*
It beats a reasoning judge	read-out − judge (two-sided)	+0.073 [0.021, 0.124]	*anon.*
Another answerer does as well	own − other (reading fixed in advance)	+0.012 [−0.018, 0.040]: no difference	*anon.*
The label-free SNR predicts where it works	Spearman over cells (lower bound > 0)	0.710 [0.436, 0.870]	*anon.*

Claim	Registered endpoint (bar)	Result	Lock
A question's spread predicts its value	mean Spearman (lower bound > 0)	0.580 [0.470, 0.649]	*anon.*
Prompt injection (AgentDojo)	within-suite AUROC, each agent (lower bound > 0.60)	0.904, 0.985	*anon.*
GPT-6-luna at its call	within-stratum AUROC (lower bound > 0.5)	0.979 [0.946, 0.998]	*anon.*
Claude Haiku 5.5 on τ^2-bench	within-tool AUROC (lower bound > 0.5)	0.704 [0.536, 0.822]	*anon.*
FORECASTING BEFORE THE CALL (Section 4.2)			
GPT-6-luna, beyond position	within-position AUROC (lower bound > 0.50)	0.923 [0.811, 0.992]	*anon.*
second fresh set (pooled with Claude Haiku 5.5)	within-position AUROC, pooled (lower bound > 0.50)	0.649 [0.570, 0.705]	*anon.*
Claude Haiku 5.5's own sampled policy	within-position AUROC (lower bound > 0.50)	0.708 [0.648, 0.765]	*anon.*
better than its self-report	own $-$ self-report (lower bound > 0)	+0.233 [0.161, 0.312]	*anon.*
on the same calls, exact contexts	own $-$ self-report (lower bound > 0)	+0.417 [0.370, 0.473]	*anon.*
better than a reasoning text monitor	own $-$ judge, within position (lower bound > 0)	+0.443 [0.368, 0.523]	*anon.*
better than another provider's text monitor	own $-$ GPT-6-luna judge, within position (lower bound > 0)	+0.295 [0.213, 0.380]	*anon.*
sampling is what matters	GPT-6-luna reading the draws $-$ reading the transcript (lower bound > 0)	+0.334 [0.235, 0.421]	*anon.*
the reader does not matter	GPT-6-luna's read $-$ Claude's read, inside ± 0.05 (fails: GPT-6-luna reads better)	+0.039 [0.005, 0.072]	*anon.*
The theory predicts another model's loss (exact contexts)	observed $-$ predicted Brier excess inside ± 0.02	+0.004 [-0.003, 0.010]	*anon.*
on GPT-6-luna	observed $-$ predicted Brier excess inside ± 0.02	+0.002 [-0.004, 0.008]	*anon.*
The theory predicts each text monitor's loss	observed $-$ predicted Brier excess, each of 3 pairs inside ± 0.02	+0.003 [-0.003, 0.009]; 0.000 [-0.007, 0.007]; -0.007 [-0.015, 0.000]	*anon.*†
the label-free gap separates the regimes	Claude gap $-$ GPT gap (lower bound > 0; fails)	-0.008 [-0.036, 0.020]	*anon.*†
a rank-based label-free check separates them	within-step agreement, GPT-6-luna $-$ Claude (lower bound > 0)	+0.127 [0.022, 0.202]	*anon.*†
with no label at all	observed $-$ predicted inside ± 0.02 (fails; re-calibrated reader also fails)	-0.226 [-0.246, -0.206]	*anon.*†
Across steps, draws add to the best text monitor	judge + draws $-$ judge (lower bound > 0)	+0.059 [0.044, 0.073]	*anon.*†
The geometry alone forecasts	within-position AUROC, each closed agent (lower bound > 0.50)	0.892 [0.853, 0.924]; 0.642 [0.539, 0.737]	*anon.*†
REPLICATION ON FRESH EPISODES (registered before any episode; Holm over the 8 fixed in advance)			
Own sampled actions beat the self-report (primary)	own $-$ self-report (lower bound > 0)	+0.405 [0.356, 0.457]	*anon.*
beat Claude's text monitor	own $-$ judge (lower bound > 0)	+0.328 [0.254, 0.401]	*anon.*
beat GPT-6-luna's text monitor	own $-$ judge (lower bound > 0)	+0.260 [0.185, 0.333]	*anon.*
forecast at all	AUROC within position (lower bound > 0.5)	0.875 [0.831, 0.913]	*anon.*
add to the best text monitor across steps	judge + draws $-$ judge (lower bound > 0)	+0.044 [0.029, 0.058]	*anon.*
predict each text monitor's loss	observed $-$ predicted, inside ± 0.02	+0.002 [-0.002, 0.007]; -0.003 [-0.010, 0.003]	*anon.*
sampling, not the reader	another reader of the draws $-$ of the transcript (lower bound > 0)	+0.302 [0.228, 0.372]	*anon.*
The geometry alone flags and forecasts	monitor; forecast (each lower bound > 0.5)	0.806 [0.777, 0.834]; 0.866 [0.820, 0.905]	*anon.*
The geometry alone forecasts on four open agents	mean AUROC (lower bound > 0.5)	0.748 [0.678, 0.815]	*anon.*†
against a probe on the stored state	geometric $-$ probe (lower > -0.10; fails)	-0.156 [-0.277, -0.042]	*anon.*†
The answers estimate a slice of the decision state	slice $-$ full state; slice $-$ random rank 8 (lower > 0, ≥ 3 of 4)	4 of 4; 4 of 4	*anon.*†

Claim	Registered endpoint (bar)	Result	Lock
the fair white box at turn start	geometric − turn-start probe (lower > -0.05; fails)	-0.024 [-0.159, 0.118]	*anon.*[†]
GPT-6-luna's own sampled policy, six templates	within-template AUROC (lower bound > 0.50)	0.681 [0.575, 0.778]	*anon.*
Open agents, within a step	pooled AUROC (lower bound > 0.5)	0.669 [0.594, 0.764]	*anon.*
The deciding fact enters at the call's step	step − earlier step (lower bound > 0; secondary)	+0.16 [0.03, 0.30]	*anon.*
A Kalman forecast beats position	forecast − position prior (lower bound > 0)	+0.21 [0.08, 0.32]	*anon.*
PREVENTING VIOLATIONS (Section 4.3)			
Self-policing, Gemma-4	relative reduction (lower bound ≥ 0.50)	0.82 [0.76, 0.88]	*anon.*
Self-policing, Claude Haiku 5.5	relative reduction, decidable (lower bound ≥ 0.40)	0.69 [0.51, 0.88]	*anon.*
The prevention law	each $\lvert\text{error}\rvert \leq 0.15$; mean ≤ 0.10	5 of 5; mean 0.035	*anon.*
Self-policing against injection	Gemma-4 reduction (lower bound ≥ 0.50)	0.84 [0.76, 0.91]	*anon.*
Pre-call gate on GPT-6-luna	pre − position at matched rate (lower bound > 0)	+0.313 [0.062, 0.543]	*anon.*
its effect predicted	$\lvert\text{predicted} - \text{observed}\rvert \leq 0.15$	0.750 vs 0.750	*anon.*
No reviewer: the agent's own read-out holds its calls	self reduction (lower bound > 0)	0.473 [0.392, 0.554]	*anon.*
more than random holds	self − random (lower bound > 0)	+0.198 [0.112, 0.290]	*anon.*
at a matched dose, fresh seeds	self − random (lower bound > 0); dose within 0.03	+0.114 [0.035, 0.200]; 0.214 vs 0.234	*anon.*
The geometry alone gates live, no reviewer	geometry gate vs none (lower > 0)	0.318 [0.220, 0.411]	*anon.*
and stacks on a policy reminder	reminder + geometry vs reminder (lower > 0)	0.490 [0.345, 0.617]	*anon.*
but not beyond random holds at its dose	random − geometry (lower > 0; fails)	+0.033 [-0.035, 0.100]	*anon.*
with the tie fixed, beyond random holds	random − geometry (lower > 0)	+0.122 [0.071, 0.177]	*anon.*
No reviewer, GPT-6-luna, six templates	self reduction (lower bound > 0)	0.353 [0.222, 0.476]	*anon.*
more than random holds	self − random (lower bound > 0)	+0.206 [0.045, 0.388]	*anon.*

Multiple tests

The family is every registered endpoint, passed or failed, of every closed-agent test whose result the abstract cites, and of the tests that bear on those claims (37 endpoints; open-agent endpoints are left out). For each, a one-sided p-value against its registered bar comes from its estimate and its 95% cluster-bootstrap interval (normal approximation, SE = interval width / 3.92; for an equivalence endpoint, the larger of its two one-sided p-values). Holm's step-down then keeps the family-wise error at 0.05. This check is exploratory: we chose it after the outcomes, and revised it after review (a first version mixed in two open-agent endpoints). Of the 37 endpoints, 26 passed as registered and 25 survive Holm; the one that passed but does not survive is GPT-6-luna's reviewer-free gate over random holds.

Endpoint	est.	bar	registered	p	Holm
Read-out at the call: Claude restarts	0.909	>0.65	pass	$< 10^{-6}$	**yes**
GPT-6-luna read-out at the call	0.979	>0.5	pass	$< 10^{-6}$	**yes**
Claude own draws − Claude text judge	0.443	>0	pass	$< 10^{-6}$	**yes**
Geometric monitor, Claude	0.818	>0.5	pass	$< 10^{-6}$	**yes**
Geometric forecast, Claude	0.892	>0.5	pass	$< 10^{-6}$	**yes**
Claude own draws − self-report, same calls	0.417	>0	pass	$< 10^{-6}$	**yes**
Draws add to Claude's judge, across steps	0.118	>0	pass	$< 10^{-6}$	**yes**
Claude reviewer-free gate: reduction	0.473	>0	pass	$< 10^{-6}$	**yes**
Draws add to position, across steps	0.125	>0	pass	$< 10^{-6}$	**yes**
Draws add to GPT-6-luna's judge, across steps	0.059	>0	pass	$< 10^{-6}$	**yes**
GPT-6-luna reads Claude's draws − reads transcript	0.334	>0	pass	$< 10^{-6}$	**yes**
Claude own sampled actions, within position	0.708	>0.5	pass	$< 10^{-6}$	**yes**
Claude own draws − GPT-6-luna text judge	0.295	>0	pass	$< 10^{-6}$	**yes**
Claude reviewer-free gate: success (non-inferiority)	0.065	>-0.05	pass	$< 10^{-6}$	**yes**
Claude own draws − self-report	0.233	>0	pass	$< 10^{-6}$	**yes**
Loss predicted, GPT-6-luna judging Claude	-0.000	$\lvert\cdot\rvert$ <0.02	pass	$< 10^{-6}$	**yes**
GPT-6-luna reviewer-free gate: reduction	0.353	>0	pass	$< 10^{-6}$	**yes**
Loss predicted, Claude's judge	0.003	$\lvert\cdot\rvert$ <0.02	pass	$< 10^{-6}$	**yes**
Claude reviewer-free gate − random	0.198	>0	pass	6.5×10^{-6}	**yes**
Self-forecast: own, within position (pooled)	0.649	>0.5	pass	7.9×10^{-6}	**yes**
GPT-6-luna pre-call read-out	0.765	>0.5	pass	3.0×10^{-5}	**yes**
Loss predicted, GPT-6-luna judging itself	-0.007	$\lvert\cdot\rvert$ <0.02	pass	4.0×10^{-4}	**yes**
Geometric forecast, GPT-6-luna	0.642	>0.5	pass	2.5×10^{-3}	**yes**
Text-policy agreement, GPT-6-luna − Claude	0.127	>0	pass	2.7×10^{-3}	**yes**
Geometric monitor, GPT-6-luna	0.816	>0.5	pass	2.9×10^{-3}	**yes**
GPT-6-luna reviewer-free gate − random	0.206	>0	pass	9.4×10^{-3}	no
GPT-6-luna reviewer-free gate: success (non-inferiority)	-0.016	>-0.05	fail	5.4×10^{-2}	no
GPT-6-luna read − Claude read, equivalence 0.05	0.039	$\lvert\cdot\rvert$ <0.05	fail	2.6×10^{-1}	no
Loss predicted, recalibrated reader, Claude's judge	-0.016	$\lvert\cdot\rvert$ <0.02	fail	2.7×10^{-1}	no
Label-free gap, Claude − GPT-6-luna	-0.008	>0	fail	7.1×10^{-1}	no
GPT-6-luna own draws − GPT-6-luna text judge	-0.049	>0	fail	7.8×10^{-1}	no
Self-forecast: own − other, within position (pooled)	-0.032	>0	fail	8.2×10^{-1}	no
Loss predicted, recalibrated reader, GPT-6-luna judging Claude	-0.046	$\lvert\cdot\rvert$ <0.02	fail	1.0×10^{0}	no
Claude own draws − Gemma-4's draws, within template|tool	-0.144	>0	fail	1.0×10^{0}	no
Self-forecast: own − other, within template|tool (pooled)	-0.190	>0	fail	1.0×10^{0}	no
Loss predicted with no label, Claude's judge	-0.226	$\lvert\cdot\rvert$ <0.02	fail	1.0×10^{0}	no
Loss predicted with no label, GPT-6-luna judging Claude	-0.214	$\lvert\cdot\rvert$ <0.02	fail	1.0×10^{0}	no

Every preregistered test

This appendix lists every preregistered test, passed or failed, in the order run. Lock commits are public (withheld here for anonymous review); details, data and causes of failure are in the repository's record.

Tally (a test passes only if it passes every primary endpoint). Method tests: 82; 38 passed every primary endpoint, 26 mixed (passed some endpoints and failed others), 14 failed, 2 stopped before any outcome, 2 pending. Trained-comparison tests: 32; 15 passed every primary endpoint, 7 mixed, 7 failed, 3 with no pass/fail bar. The seven earlier tests of the observable state in chat (six passed, one mixed) are listed but not counted. Two method tests compare trained monitors; the table lists them with the trained comparisons. No correction for multiple tests is applied. A lock is the public commit that published the preregistration before its test ran; where the record names no commit, the table gives the lock date (2026, PDT). Brackets are 95% intervals.

Test	Primary endpoint and result	Outcome	Lock
Geometry			
Observable state, open chat models	Three gains, after one recorded deviation: +0.120 [0.115, 0.126]; +0.057 [0.045, 0.073]; +0.359 [0.310, 0.415]	pass	2026-10-15
Observable state, Othello	R^2 0.967 [0.967, 0.968] > 0.90; gain in R^2 +0.094 [0.086, 0.105]	pass	2026-10-15
Open-model readouts transfer to Claude Haiku 5.5	0.993 [0.983, 0.999]; 0.997 [0.990, 1.000]; 0.935 [0.868, 0.982]	pass	2026-10-15
Answers vs input text, 8 open models and Claude; exact recovery	Answers beat text on 8/8 models (lower bounds $\geq$ 0.141); Claude split-half 0.950 [0.939, 0.960], vs text +0.438 [0.391, 0.485]; exact Gram recovery, Pearson 1.0	pass	*anon.*
Yes/no answers vs the model's own replies	Exact answers beat 8 replies on 8/8 models (+0.16 to +0.48); sampled answers beat 1 reply on 7/8; replies saturate on 8/8	pass	*anon.*
The geometry is shared across models	No own-model excess, +0.002 [−0.001, +0.005] (equivalence ±0.02); an open model's hidden geometry is the better stand-in on 6/7	pass	*anon.*
What each API channel reveals	Answers show no own-model excess (+0.001); log-probabilities leak most, +0.221 [0.211, 0.230], and identify the model on 7/8	pass	*anon.*
Estimated state vs the true hidden state	Filtered − raw answers +0.019 [0.001, 0.038]; forecast 0.116 [0.078, 0.153] (mostly position, exploratory)	pass	*anon.*

Test	Primary endpoint and result	Outcome	Lock
Memory of the state and the filter's noise	Lag-1 autocorrelation 0.132 [0.090, 0.171]; decay 0.119 [0.075, 0.155]; variance 0.31 [0.29, 0.34] of the nominal noise	pass	*anon.*
Triangulation: a closed model's own hidden geometry	Leave-one-out error 0.061 [0.051, 0.070]; 7 of 8 inside the interval; Claude Haiku 5.5 0.495 [0.437, 0.554]	pass	*anon.*
Does the monitor come from the reconstructed state?	20% [13%, 27%] of the read-out's AUROC runs through the hidden violation coordinate (pass); within template and label 0.044 [−0.008, 0.115] (fail)	mixed	*anon.*
With dither, does the read-out follow the hidden state?	Within template × tool × step × label −0.077 [−0.250, 0.195]; dither − repeats −0.041 [−0.274, 0.201]	fail	*anon.*
Triangulation at the agent's decision	Mean error 0.045 [0.031, 0.065] (pass); inside the interval for 1 of 4 (fail); Claude Haiku 5.5 0.103 [0.085, 0.122] (pass)	mixed	*anon.*
A broader battery (32 questions) at the decision	Spearman with the decision-state geometry, 32 − 8 questions, +0.120 [0.084, 0.156] (pass); 32 questions 0.180 [0.155, 0.204], bar 0.2 (fail)	mixed	2026-10-23
Monitoring at the call			
The agent's own answers, training-free	8 answers − 1 answer +0.036 [0.006, 0.064]; − trained probe +0.045 [−0.017, 0.135] (margin −0.05); Claude restarts 0.909 [0.858, 0.952]	pass	*anon.*
Monitoring by the reconstructed geometry alone	Fisher–Rao distance from the typical state, no label or sign, within stratum: Claude 0.818 [0.792, 0.844], GPT-6-luna 0.816 [0.537, 0.986]	pass	*anon.*[†]
Own answers or another model's	Own − other, four open agents, +0.012 [−0.018, 0.040]: no detectable difference	fail	*anon.*
Read-out on prompt injection (AgentDojo)	Within-suite AUROC: Gemma-4 0.985 [0.964, 0.998]; gpt-oss-20b 0.904 [0.872, 0.932] (bar 0.60)	pass	*anon.*
A closed agent's own answers, tau2-bench	Claude Haiku 5.5, within-tool AUROC 0.704 [0.536, 0.822] (bar 0.50)	pass	*anon.*
Own conversation or outside reviewer, same closed model	Own − outside +0.043 [−0.136, 0.226]: no detectable difference	fail	*anon.*
Questionnaire or reasoning judge	Two-sided: read-out − Gemma-4 reasoning judge, four open agents, +0.073 [0.021, 0.123] (questionnaire better)	pass	*anon.*
An open model answers about a closed agent	Gemma-4 about Claude's tau2-bench writes, within tool 0.989 [0.954, 1.000] (bar 0.50)	pass	*anon.*
One universal questionnaire, two benchmarks	CAPE-Bench 0.890 [0.818, 0.950]; tau2-bench 0.935 [0.841, 0.985] (both pass); reported: universal − tailored, no detectable difference on CAPE-Bench	pass	*anon.*
Label-free theory: which questions monitor	Spearman of SNR with AUROC 0.745 [0.608, 0.779] (pass); SNR weights − equal +0.008 [−0.177, 0.032] (no detectable difference)	mixed	*anon.*
A closed agent's own answers, with reasoning	Within-tool AUROC 0.568 [0.454, 0.771] (bar 0.50); on − off −0.051 [−0.246, 0.150]	fail	*anon.*
Label-free theory on the agent's own answers	Spearman of SNR with AUROC 0.580 [0.470, 0.648] (pass); SNR weighting lowers the read-out, −0.068 [−0.142, −0.047]	pass	*anon.*
A closed model reviews its own calls, with reasoning	Within-tool AUROC 0.918 [0.730, 0.990] (bar 0.50); − reviewer without reasoning +0.262 [0.007, 0.478]	pass	*anon.*
A label-free signal says where the read-out works	Spearman 0.710 [0.436, 0.870]; cells with SNR >0 − SNR ≤ 0 +0.282 [0.169, 0.406]	pass	*anon.*
A closed agent's answers, dithered	Dither − 4 answers +0.015 [−0.089, 0.104]; dither − 16 repeats +0.034 [−0.088, 0.157]	fail	*anon.*
A second closed agent, at its call and before it	GPT-6-luna, within template × tool: at the call 0.979 [0.946, 0.998]; before it 0.765 [0.650, 0.909] (bar 0.50)	pass	*anon.*
Reading along the directions the call can violate	Relevant − full read-out −0.014 [−0.059, 0.038] (dithered); +0.008 [−0.021, 0.026] (earlier answers)	fail	*anon.*
Forecasting before the call			
State tomography and forecasting	Forecast − position prior +0.013 [−0.115, 0.155]; filtered − raw −0.022 [−0.073, 0.030]; the linear dynamics do not fit	fail	*anon.*
State tomography, three more open agents	Forecast − position prior +0.209 [0.080, 0.317] (pass); filtered − raw +0.018 [−0.021, 0.062]	mixed	*anon.*
The state before a call, beyond position	Within-step AUROC 0.669 [0.594, 0.764] (pass); same-step agreement with the true state 0.072 [−0.004, 0.144] (fail)	mixed	*anon.*
Repeats or dither: is the within-step effect measurement-limited?	16 dithered − 4 answers +0.082 [0.001, 0.147] (pass); dither − 16 repeats +0.006 [−0.064, 0.066] (fail); true state +0.053 [−0.014, 0.129] (fail)	mixed	*anon.*
Early warning and the horizon of the state	Early warning 0.498 [0.405, 0.585] (fail); before call t − before call $t-1$ +0.163 [0.029, 0.302] (pass)	mixed	*anon.*
A closed agent's state before its call, tau2-bench	Claude Haiku 5.5, pre-call within-tool AUROC 0.536 [0.434, 0.682] (bar 0.50); dither − repeats +0.127 [−0.045, 0.253]	fail	*anon.*
Own forecast or another model's	GPT-6-luna own − other +0.111 [0.041, 0.185] (pass); three open agents −0.030 [−0.159, 0.082] (fail)	mixed	2026-10-23
Is the own advantage specific to forecasting?	(Own − other) before the call − at the call, three open agents, −0.045 [−0.178, 0.082]	fail	2026-10-23
Own or other forecast, two closed agents	Own − other −0.190 [−0.238, −0.142] (fail); own within position 0.649 [0.570, 0.705] (pass); own − other within position −0.032 [−0.100, 0.038] (fail)	mixed	2026-10-23
The forecast lives in the agent's policy	Own draws within position 0.708 [0.648, 0.765] (pass); own − other −0.144 [−0.195, −0.094] (fail); own − self-report +0.233 [0.161, 0.312] (pass)	mixed	*anon.*
The theory predicts the other model's loss in size	Observed − predicted Brier excess −0.046 [−0.062, −0.032], bar ±0.02 (fail); policy gap 0.095 [0.083, 0.107] (pass)	mixed	*anon.*
The policy forecast with the exact context	Another model's observed − predicted Brier excess +0.004 [−0.003, 0.010] (primary, bar ±0.02); calibration +0.004 [−0.005, 0.012]; own within position 0.891 [0.855, 0.927]; policy gap 0.111 [0.095, 0.128]	pass	*anon.*
The policy forecast on a second closed agent, six templates	GPT-6-luna own draws within template and tool 0.681 [0.575, 0.778], within position 0.729 [0.615, 0.824] (pass); calibration +0.015 [0.008, 0.023] (pass); another model's loss, observed − predicted +0.002 [−0.004, 0.008] (pass, bar ±0.02)	pass	*anon.*
The policy forecast on an external benchmark (τ^2-bench)	Claude Haiku 5.5 own draws within tool 0.605 [0.444, 0.835], bar 0.5	fail	*anon.*

Test	Primary endpoint and result	Outcome	Lock
A reasoning text monitor vs the agent's own policy	Own draws − the same model reasoning over the transcript, within position, +0.443 [0.368, 0.523] (own 0.891, judge 0.448)	pass	*anon.*
Each reader's loss predicted by the policy gap	Mean over three more open readers of observed − predicted, bar ±0.02	pending	*anon.*
A reasoning text monitor vs the agent's own policy, GPT-6-luna	Own draws − GPT-6-luna reasoning over the transcript, within position, −0.049 [−0.178, 0.071] (own 0.729, judge 0.777)	fail	*anon.*
A different provider's text monitor on Claude's transcripts	Own draws − GPT-6-luna reasoning over Claude's transcripts, within position, +0.295 [0.213, 0.380] (judge 0.596)	pass	*anon.*
Forecasting by the reconstructed geometry alone	Mean Fisher–Rao distance of the sampled next actions from the typical state, within position: Claude 0.892 [0.853, 0.924], GPT-6-luna 0.642 [0.539, 0.737]	pass	*anon.*†
Another model reads Claude's sampled actions	GPT-6-luna reading Claude's draws − reading its transcript, within position, +0.334 [0.235, 0.421] (pass); − Claude's own reads +0.039 [0.005, 0.072], outside ±0.05 (fail; GPT-6-luna reads better)	mixed	*anon.*
The theory predicts a text monitor's loss	Observed − predicted Brier excess, inside ±0.02 for each pair: Claude's judge +0.003 [−0.003, 0.009], GPT-6-luna on Claude 0.000 [−0.007, 0.007], GPT-6-luna on itself −0.007 [−0.015, 0.000] (pass); label-free gap Claude − GPT −0.008 [−0.036, 0.020] (fail)	mixed	*anon.*†
Says vs does on the same calls, exact contexts	Claude's sampled actions − its self-report, within position, +0.417 [0.370, 0.473] (self-report 0.473 [0.436, 0.503])	pass	*anon.*
Across steps, sampled actions add to text	Mean rank with Claude's draws − GPT-6-luna's judge alone, within tool, +0.059 [0.044, 0.073]; with Claude's judge +0.118 [0.093, 0.144]; with position +0.125 [0.094, 0.155]	pass	*anon.*†
The loss predicted with no label at all	GPT-6-luna's probabilities in place of the checker: observed − predicted −0.226 [−0.246, −0.206] (calibration −0.195)	fail	*anon.*†
with a recalibrated reader	Recalibrated on 200 labelled draws from other scenarios: −0.016 [−0.030, −0.004] (Claude's judge), −0.046 (GPT-6-luna's)	fail	*anon.*†
Label-free rank agreement separates the regimes	Within-step Spearman of text monitor and sampled policy, GPT-6-luna − Claude, +0.127 [0.022, 0.202] (constant-stratum check fixed after the lock; before, [0.032, 0.199])	pass	*anon.*†
Replication on fresh episodes of the forecasting results	492 new Claude episodes, 8 endpoints, Holm fixed in advance: own − self-report +0.405 [0.356, 0.457] (primary); own − Claude's judge +0.328 [0.254, 0.401]; own − GPT-6-luna's judge +0.260 [0.185, 0.333]; across-steps lift +0.044 [0.029, 0.058]; own 0.875 [0.831, 0.913]; loss prediction +0.002 and −0.003 (bar ±0.02); another reader of the draws − of the transcript +0.302 [0.228, 0.372]; all 8 survive Holm	pass	*anon.*
Replication of the geometry-only monitor and forecast	Monitor 0.806 [0.777, 0.834]; forecast 0.866 [0.820, 0.905] (bars: lower > 0.5)	pass	*anon.*
Prevention at a matched dose, fresh seeds	Self − random reduction +0.114 [0.035, 0.200] (primary); holds per call 0.214 vs 0.234 (matched); violating episodes 67.9% / 35.2% / 42.9% (none / self / random); success +0.081 [0.039, 0.130]; Holm passes	pass	*anon.*
Says vs does on four open agents, many task families	Mean over agents of sampled actions − self-report, within template, tool and step: +0.095 [0.001, 0.164] (primary); sampled actions 0.765 [0.693, 0.827]; higher on 4 of 4 agents (Muse-Glimmer rerun under vLLM 0.30, recorded deviation)	pass	*anon.*
Geometry-only forecast on four open agents; against a white-box probe	Mean geometric AUROC 0.748 [0.678, 0.815] (pass; 3 of 4 agents above chance; Muse-Glimmer 0.556); geometric − probe on the stored (after-reasoning) state −0.156 [−0.277, −0.042] (fail, bar −0.10)	mixed	*anon.*†
The fair white box; the slice the answers estimate	Turn-start probe vs API geometry on Gemma-4: −0.024 [−0.159, 0.118] (fail, non-inferiority bar −0.05; turn-start − after-reasoning probe −0.198); answers vs the slice they pull back 0.36–0.60 vs full state 0.01–0.16 and random rank-8 0.02–0.10 (pass, 4 of 4); the 32-question geometry alone as a monitor 0.50 [0.36, 0.65] (fail; the 8 policy questions' geometry 0.58–0.96, reported)	mixed	*anon.*†
A geometry-only live gate, stacked on a policy reminder	Violating episodes 65.9% (none), 44.9% (geometry gate), 21.1% (reminder), 10.8% (both); reminder + geometry vs reminder 0.490 [0.345, 0.617] (primary); geometry vs none 0.318 [0.220, 0.411]; success +0.209 [0.136, 0.297]; all survive Holm	pass	*anon.*
A closed model's geometry on fresh contexts	Claude's answers vs open hidden geometries 0.10–0.48; its own replies (embedded) stronger on 6 of 8 models; answers win on 1 of 8 (bar: 6)	fail	*anon.*
The fair white box, powered	Geometric − turn-start probe on 865 Gemma-4 units, lower > −0.05	pending	*anon.*
The geometry-only gate against random holds, matched dose	Violating episodes 44.9% (geometry) vs 48.2% (random) at 0.172 vs 0.170 holds per call; random − geometry +0.033 [−0.035, 0.100] (fail); dose matched (pass)	mixed	*anon.*
The geometry-only gate against random holds, tie fixed	Violating episodes 20.1% (geometry, at or above the cut) vs 32.3% (random) at 0.32 vs 0.30 holds per call; random − geometry +0.122 [0.071, 0.177] (pass); dose matched (pass)	pass	*anon.*
A closed agent's pre-call state beyond position, confirmed	GPT-6-luna, fresh episodes: within position 0.923 [0.811, 0.992]; within template × tool 0.840 [0.732, 0.946] (bar 0.50)	pass	2026-10-22
Prevention			
Live self-policing through the agent's API	Gate reduction 0.82 [0.76, 0.88] (pass); resampling 0.404 reviews per episode, limit 0.40 (fail); success +10.2 [+6.6, +14.3] points (pass)	mixed	*anon.*
A closed agent polices itself	Decidable: reduction 0.69 [0.51, 0.88]; unchecked fact: 0.07 [0.00, 0.17], as predicted; success −0.3 [−2.1, +1.6] points	pass	*anon.*
The prevention law across agents, first design	Stopped: after 300 Qwen3.8 episodes neither arm had a violation; replaced by the next test	stopped	*anon.*
The prevention law across agents	Law within 0.15 (fail: Muse-Glimmer 0.23 [0.13, 0.35] vs 0.722); Muse > Qwen +0.29 [0.10, 0.50] (pass); success (fail)	mixed	*anon.*
The composition-weighted law	Within 0.15 for 5/5 agents; mean absolute error 0.035 (≤ 0.10)	pass	*anon.*
Self-policing against prompt injection	Gemma-4 reduction 0.843 [0.756, 0.912] (pass); law within 0.15: gpt-oss pass, Gemma-4 miss 0.157 (fail); utility pass	mixed	*anon.*
Live prevention by the estimated state	Stopped before any episode outcome: the dynamics were unstable and there was no position control; replaced	stopped	*anon.*
Live pre-call gate against a position gate	Gemma-4: reduction −0.04 [−0.30, 0.16] (fail); pre − position −0.24 [−0.50, −0.03] (fail); success +0.023 [−0.007, 0.053] (pass)	mixed	*anon.*
Live pre-call gate on a closed agent	GPT-6-luna: reduction 0.636 [0.250, 0.875] (pass); pre − position +0.227 [−0.190, 0.625] (fail); success +0.012 [−0.083, 0.107] (fail)	mixed	2026-10-22

Test	Primary endpoint and result	Outcome	Lock
Pre-call gate against a position gate, matched review rate	Pre − position +0.313 [0.062, 0.543]; reduction 0.750 [0.605, 0.863]; success 0.000 [−0.048, 0.048]	pass	2026-10-22
The prevention law for pre-call gates	Pre-call gate: predicted 0.750, observed 0.750 [0.605, 0.863] (pass); position gate: 0.167 vs 0.438, error 0.271 (fail)	mixed	2026-10-22
Prevention with no reviewer: the agent re-checks a held call	Reduction 0.473 [0.392, 0.554]; self − random holds +0.198 [0.112, 0.290]; success +0.065 [0.033, 0.102]	pass	*anon.*
Forecast, then a policy reminder, no reviewer	Reduction 0.778 [0.709, 0.841] (pass); forecast − random timing +0.048 [−0.012, 0.106] (fail); success +0.266 [0.171, 0.372] (pass)	mixed	*anon.*
No reviewer, second closed agent (six templates)	GPT-6-luna reduction 0.353 [0.222, 0.476] (pass); self − random +0.206 [0.045, 0.388] (pass); success −0.016 [−0.056, 0.028], bar −0.05 (fail)	mixed	*anon.*
Trained comparisons			
Resampling vs a white-box probe	Resampling − probe +0.063 [0.033, 0.094] (pass); observable state − probe −0.021 [−0.054, 0.009] (fail)	mixed	2026-10-15
When the internals are read	Internals at the decision − resampling at the turn start +0.075 [0.044, 0.106]; all three endpoints pass	pass	2026-10-16
Resampling with a deployable judge	Judge-scored resampling − turn-start probe −0.065 [−0.115, −0.019]	fail	2026-10-16
Trained reader vs the agent's internals	Reader − internals +0.015 [0.007, 0.026]; reader − judge +0.085 [0.052, 0.127]	pass	2026-10-16
Unseen task families	Reader − internals −0.007 [−0.014, −0.0001] (margin −0.02)	pass	2026-10-16
Another open agent	Gemma-trained reader on gpt-oss 0.915 vs judge 0.833, +0.082 [0.043, 0.129]	pass	2026-10-16
AgentDojo, no transfer training	Reader − zero-shot judge +0.249 [0.136, 0.369]	pass	2026-10-16
Claude Haiku 5.5 as agent, after a reveal	Not tested: 0 unsafe units under the post-reveal rule	no bar	2026-10-16
Closed agent: Claude Haiku 5.5	Reader 0.991 vs zero-shot judge 0.633, +0.360 [0.302, 0.413]	pass	2026-10-16
Reader vs TF-IDF	Unseen families +0.019 [0.007, 0.032] (pass); Claude −0.006 [−0.027, +0.007] (fail)	mixed	2026-10-16
Claude Haiku 5.5 on unseen task families	Reader − judge: pooled +0.342 [0.311, 0.371]; within +0.318 [0.265, 0.374]	pass	2026-10-16
Closed agent: GPT-6-luna	Reader − judge +0.217 [0.174, 0.265]; − tool identity +0.141 [0.111, 0.171]; within tool +0.039 [0.015, 0.066]	pass	2026-10-17
Labels for a new policy	40 − 0 labels: 0.0 (fail: 18/18 caught with 0 labels); catch at 40 $\geq$ 0.5 (pass)	mixed	2026-10-17
AgentDojo with a defense, closed agent	Not tested: 4 injected calls (minimum 10)	no bar	2026-10-17
AgentDojo without the defense, closed agent	Reader − zero-shot judge +0.052 [−0.026, +0.130]	fail	*anon.*
Open-agent labels for a closed agent	Reader − judge +0.136 [0.078, 0.191]; − no labels +0.081 [0.044, 0.121]	pass	2026-10-17
Replication of the hidden state	Replica − text, variance explained +0.079 [0.075, 0.083]; probe on replica − true state +0.003 [−0.002, +0.007]	pass	2026-10-17
Live prevention, GPT-6-luna	Violating episodes 6.6% → 0.2%, −0.064 [−0.088, −0.044]	pass	*anon.*
Live prevention, Claude Haiku 5.5	Violating episodes 6.7% → 0.2%, −0.064 [−0.087, −0.044]	pass	2026-10-17
Replication, three architectures	R^2 over text: +0.102 [0.093, 0.110]; +0.101 [0.091, 0.111]; +0.061 [0.051, 0.071]	pass	2026-10-17
Hold for review	Violating episodes 6.7% → 0.0%, −0.067 [−0.091, −0.044] (pass); goals −0.020 [−0.051, +0.007] (fail)	mixed	2026-10-17
tau2-bench, Claude Haiku 5.5	Not tested: 9 violations, fewer than the minimum of 15	no bar	2026-10-17
Browser-size readers	Reader − TF-IDF +0.270 [0.134, 0.451] (pass); − 8B reader −0.097 [−0.115, −0.079] (fail)	mixed	2026-10-17
tau2-bench, two closed agents	TF-IDF − tool +0.487 [0.428, 0.500] (pass); reader − tool +0.238 [−0.053, 0.431] (fail)	mixed	*anon.*
tau2-bench, unseen tasks	Reader − TF-IDF +0.017 [−0.514, 0.539]; reader − tool +0.035 [−0.230, 0.259]	fail	*anon.*
tau2-bench, reasoning judge	Judge − tool +0.304 [−0.084, 0.427]; judge − reader +0.268 [−0.074, 0.511]	fail	*anon.*
Reasoning judge, frontier agents	Judge − tool +0.266 [0.151, 0.338] (pass); judge − reader +0.120 [−0.047, 0.274] (fail)	mixed	*anon.*
Monitor-aware agent	Caught, aware − plain +0.136 [+0.050, +0.225]; aware 97.1% [92.9%, 100%]	pass	*anon.*
Adaptive attacker	Caught 96.3% → 29.1% [21.6%, 36.6%]; reader − TF-IDF −0.052 [−0.149, +0.037]	fail	*anon.*
Canonical-action monitor	Caught after attack 50.8% [42.5%, 59.0%]; clean catch −0.410 [−0.493, −0.311]	fail	*anon.*
Training against the attack	Caught after attack 59.7% [51.5%, 67.9%]; clean catch −0.037 [−0.078, +0.016]	fail	*anon.*
Cut set again after a prompt change	Violating episodes 6.5% → 0.0%, −0.065 [−0.090, −0.043]; escalated 4.73% [3.49%, 6.10%]	pass	*anon.*
Reading the agent's reasoning, trained monitor	In-agent gain −0.005 [−0.018, +0.013] (fail); reasoning hurts transfer −0.033 [−0.046, −0.017] (pass); 2 of 4 pass	mixed	*anon.*
Recreating the decision state for a white-box probe	Recreation − true state +0.063 [0.026, 0.122]; other agents' coordinates +0.009 [−0.050, 0.097]; Claude restarts 1.000	pass	*anon.*
Other			
Claude Haiku 5.5 refusals vs moderation	0.923 [0.884, 0.957]; grey box − moderation +0.176 [0.060, 0.287]	pass	2026-10-16
Claude Haiku 5.5, one-call grey box	Grey box − 3-level verdict +0.142 [0.081, 0.203]; alarm AP +0.230 [0.161, 0.307]	pass	2026-10-16
Claude Haiku 5.5, transcript readout	Grey box − moderation +0.141 [0.082, 0.199]; compliance-risk AP +0.115 [0.076, 0.179]	pass	2026-10-16
Claude Haiku 5.5 vs graded moderators	Alarm precision +0.261 [0.215, 0.311] (pass); vs graded Qwen3Guard +0.018 [−0.023, 0.062] (fail); multi-turn 0.894 vs 0.906 (fail)	mixed	2026-10-16